# Bankers on the Move: Relationship Capital in Credit Markets *

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**Abstract**

Bank loan officers accumulate information and build personal relationships that can shape credit allocation. Using Italian administrative data, we directly measure these relationships by linking bank managers to their portfolios of client firms and tracking managers as they move across banks. When a manager switches employers, firms in her prior portfolio become 3.7 percentage points more likely to establish a lending relationship with her new bank four years later, more than three times the baseline probability. Manager portability lowers firms' switching costs without generating adverse selection. Firms that follow their managers obtain lower interest rates than on their previous loans, while paying rates similar to those of other new borrowers at the same bank. Moreover, managers transfer valuable information across banks: loans generated through these relationships are less likely to become nonperforming than other new loans.

**Keywords:** Banking, Credit allocation, Human capital, Manager mobility

**JEL Codes:** G21, G32, J24, J62

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# 1 Introduction

The allocation of human capital within the financial sector has long attracted attention in both public debate and academic research (Philippon and Reshef, 2012; Boustanifar et al., 2018). Theories of financial intermediation posit that bank managers who screen and monitor borrowers accumulate valuable information about their client base (Stiglitz and Weiss, 1981). Consequently, banks design incentive-based contracts to retain those managers (Bénabou and Tirole, 2016) and to keep the information they possess within the organization (Stein, 2002). This accumulated knowledge can be crucial for firms: those maintaining close relationships with their bank managers enjoy better access to credit (Fisman et al., 2017) and are more resilient to economic shocks (Bolton et al., 2016). Yet despite this central role of bank managers, we know little about how their mobility — and the portability of the information they hold — affects credit allocation.

We present evidence that capital flows are influenced by worker flows in the banking sector — in other words, that credit follows people. Our results show that when a manager moves to a new bank, a subset of her clients follows her and obtains better credit terms than before. Banks also benefit from these manager-induced client movements: the loans of firms that follow default less frequently than those of other new borrowers. We exploit a novel combination of credit registry and financial sector social security data for Italy, the third largest economy in the European Union. For the period between 2009 and 2018, we are able to identify not only firms' credit relationships but also the managers responsible for loan issuance and monitoring. We use manager transitions across the banking sector as a source of variation in the probability that a firm establishes a new credit relationship. Specifically, we test whether this probability increases when a bank hires a manager who previously oversaw that firm's credit at another bank. To mitigate concerns that a manager's decision to move might be endogenous, we conduct a complementary analysis focusing on managerial moves triggered by branch closures.

Theoretically, the portability of lending relationships can have an ambiguous impact on credit allocation. On the one hand, labor mobility may facilitate the diffusion of borrower information across banks, allowing firms to obtain more favourable credit terms at their new institution (Fisman et al., 2017). On the other hand, if lending relationships entail private benefits for bank managers, portability may induce negative selection: new relationships could attract firms that were previously favored for non-contractible reasons (Haselmann et al., 2018), and whose risk would not be

adequately priced by the connected manager (Beraldi, 2025).

Our data and setting allow us not only to measure the portability of credit relationships but also to assess whether it is the bright or the dark side of personal connections that prevails (Bandiera et al., 2009). The Italian credit registry covers the universe of loans granted to domestic firms, providing detailed information on loan amounts, applications, interest rates, and loan performance (Malgieri and Citino, 2025; De Marco et al., 2023; Gobbi and Sette, 2014). By merging these data with firm balance sheet information from Cerved, such as Z-score riskiness (Rodano et al., 2018), we can examine how managerial knowledge shapes banks' selection processes. Crucially, each loan record identifies the originating bank branch, enabling us to link the credit registry to administrative social security records that trace the complete career histories of financial-sector employees, with workplaces identified at the branch level. This feature enables us to construct portfolios of client firms for each bank manager and to track their movements across banks over time.

On the institutional side, Italian firms typically maintain multiple banking relationships (Barone et al., 2024; Kosekova et al., 2024), which allows us to absorb firm-level credit demand and bank-level credit supply with fixed effects, following Khwaja and Mian (2008). Our estimates therefore capture the firm-bank match itself rather than aggregate demand or supply. Beginning in the 1990s and continuing through the 2000s and 2010s, the Italian banking sector underwent substantial consolidation, as many local banks merged into a few national champions (Focarelli and Panetta, 2003). This process led to the reorganization of branch networks and, in some cases, to the relocation or dismissal of bank managers. At the same time, Italian labor laws fostered both employment stability through national collective agreements that protect insiders and interbank mobility, since non-compete clauses apply only to personal bankers and not to employees engaged in lending activities.

The main finding of the paper is that credit relationships are portable across banks. Four years after a manager moves, the probability of a firm in her portfolio establishing a credit relationship with her new bank rises from a baseline of 1.3% to 5%. This impact is sizable: the 3.7 percentage point increase represents a nearly fourfold rise in the likelihood of matching. Our results hold after controlling for credit demand shocks, that affect a firm's likelihood of applying for a loan, credit supply shocks, that influence a bank's overall lending propensity, and time-invariant characteristics of the firm-bank relationship — such as geographical proximity or sectorial specialization — that affect their probability of matching. We argue that borrowers do not influence

a manager's decision to leave her former bank or to move to a specific new bank. To further mitigate endogeneity concerns, we conduct a separate analysis focusing on cases in which the manager's move is triggered by a branch closure. In these cases, the increase in the probability of a firm following its manager is even larger — rising by about 20 percentage points from the same baseline of 1.3%, before settling at about 10 percentage points above it. We interpret this stronger effect as reflecting the fact that branch closures increase firms' likelihood of separating from their former bank, thereby raising the overall probability of reestablishing a relationship with their former manager at her new institution.

Our findings suggest that managers primarily reduce search frictions for their client firms. Loan application data allow us to deeply investigate the increase in matching probability by looking at two components: the probability that a firm applies for credit (reflecting its willingness to follow the manager to her new bank) and the probability that its request is approved (reflecting the manager's influence on screening decisions). We find that both probabilities rise after a manager's move, with the increase in applications accounting for most of the effect. We interpret this pattern as evidence that managers facilitate firms' access to credit mainly by reducing search frictions rather than by altering screening outcomes. Additional evidence supports this mechanism. Firms' Z-score riskiness does not predict the probability of following the manager, suggesting the absence of adverse selection. The portability effect is instead strongest among younger and smaller firms, which are typically more reliant on soft information (Huber, 2021) and benefit most from the manager's ability to reduce search frictions.

Our second set of results examines contract terms and loan performance to assess how manager portability affects rent sharing between firms and banks. Firms that follow their manager pay interest rates that are 0.5 percentage points lower on their new loans compared with any other relationships they hold, and 0.9 percentage points lower than what they were paying at the manager's former bank. The lower interest rate serves as an incentive for firms to move their credit relationship but does not reflect preferential treatment. Indeed, when we compare firms that follow their manager with other new borrowers at the manager's new bank, the interest rates are virtually identical. Together, these results indicate that manager transitions allow firms to match with banks that can offer them better credit conditions. These conditions reflect market fundamentals, but switching costs — reduced by the manager's move — typically prevent firms from realizing such gains in the absence of

mobility. We then ask whether banks also benefit from these new relationships. We find that firms following their manager are 4 percentage points less likely to report a non-performing loan (NPL) on their loans, compared with other new borrowers at the same bank. These results remain robust after controlling for observable firm characteristics, including Z-score riskiness. These findings show that the transfer of soft information through managerial mobility benefits not only firms but also banks, by improving their ability to screen and monitor newly acquired borrowers.

Taken together, our findings provide direct evidence that bank managers generate and utilize soft information to allocate credit. This information is portable and benefits both firms and banks. The strength of portability, however, varies systematically across contexts. It is greater when managers move from larger to smaller banks, where hard information and standardized lending practices play a lesser role. It is also stronger for managers coming from smaller branches and weaker for those moving to headquarters, where tasks are less client-facing and managers are no longer directly involved in loan origination and monitoring. More experienced managers are more successful in transferring their client portfolios, likely because they have accumulated greater trust and reputation among their clients. Overall, the pattern of heterogeneity we uncover is consistent with a mechanism in which personal relationships and soft information shape the persistence of credit ties.

In terms of methods, our analysis employs a generalized difference-in-differences framework: for the same firm and year, we compare the probability of a credit relationship with the bank its former manager has just joined against that with all other banks in the same province, before and after the move. Under the parallel trends assumption, this variation allows us to estimate the causal effect of a manager's move on the likelihood that a firm establishes a new credit relationship with the manager's new bank. Our approach addresses several potential confounding factors. First, we control for time-invariant characteristics that influence the matching probability between a firm and a bank, such as industry specialization (Paravisini et al., 2023) or geographical proximity (Nguyen, 2019), through firm-bank fixed effects. Second, we absorb credit supply shocks that affect a bank's overall lending propensity using bank-time fixed effects. Third, we account for credit demand shocks that influence a firm's propensity to apply for loans through firm-time fixed effects, following Khwaja and Mian (2008). Finally, to mitigate concerns that a manager's decision to leave her branch may be endogenous to firms' credit choices, we conduct a separate analysis focusing on moves triggered by branch closures — events that are exogenous to firms'

decisions to shift their credit relationships.

Our event studies are estimated using the Sun and Abraham (2021) estimator, which accounts for cohort-specific treatment effects in settings with staggered treatment timing (De Chaisemartin and D'Haultfœuille, 2020; Borusyak et al., 2024). We conduct several additional tests to verify the robustness of our findings. First, since our baseline specification focuses on managers working in small branches, we verify that our results are not sensitive to the specific definition of branch size. Varying the branch size cutoff from 50 to 100 and 150 client firms yields consistent estimates. Second, we aggregate observations at the provincial level to address the concern that a firm might have had a pre-existing credit relationship with the manager's new bank through another branch in the same province. The results remain unchanged, confirming that our findings are not driven by such cases. Finally, we re-estimate our regressions using a Poisson specification, obtaining results fully consistent with those from the baseline linear probability model.

We provide two benchmarks to gauge the magnitude of our estimates. First, the existing literature on managerial mobility has so far been limited to movements within a single organization, typically loan officer rotations within the same banking group (Hertzberg et al., 2010). Our setting allows us to measure the portability of managerial information both across and within banks. We find that the magnitudes of the two effects are comparable, although the dynamics differ: between-bank portability builds up rapidly after the manager's move, while within-bank portability takes longer to materialize, possibly reflecting organizational frictions that slow client reallocation across branches. Second, we benchmark our estimates against client portability in a different context, such as sales managers studied by Patault and Lenoir (2024). Their effects are roughly an order of magnitude smaller than ours. We interpret this difference as evidence of the distinctive nature of banking relationships, where soft information and personal connections play a defining role.

Our paper contributes to three strands of literature. The first concerns bank branching and relationship lending, which — since the seminal work of Stiglitz and Weiss (1981) — has examined how soft information in banking relationships influences credit allocation (Amberg and Becker, 2024; Babina et al., 2022; Fisman et al., 2017) and loan pricing (Beraldi, 2025). Scholars have typically measured soft information indirectly, using proxies such as geographical proximity (Bonfim et al., 2021; Degryse and Ongena, 2005; Petersen and Rajan, 2002), cultural homophily (Frame et al., 2025; Fisman et al., 2017; Cornell and Welch, 1996), or the effects of branch closures

on lending patterns (Nguyen, 2019; Duquerroy et al., 2022). We contribute to this literature by bringing methods from labor and trade economics (Patault and Lenoir, 2024; Kramarz and Skans, 2014) to the study of banking relationships. Specifically, we combine labor market data with credit registry information to construct detailed portfolios of client firms for each bank manager and follow their evolution across the entire financial system. This approach not only allows us to provide a more precise measure of soft information, since we can track individual relationships, but also enables us to measure portability beyond the boundaries of a single bank and assess its implications for credit allocation in the financial system as a whole.

A second strand of literature explores the intersection between labor market dynamics and finance. Acabbi et al. (2024) and Jasova et al. (2021) combine workforce administrative data with credit registry information to examine how labor adjusts to credit and monetary shocks. We contribute to this line of research by studying the reverse channel — how the reallocation of labor within the financial sector affects credit allocation. Philippon and Reshef (2012) document the evolution of wage premia in the financial industry, followed by studies analyzing their drivers (Bell and Van Reenen, 2014) and their implications for talent allocation and "brain drain" (Boustanifar et al., 2018; Böhm et al., 2023). Gao et al. (2024b) develop a career model for bank employees, showing that much of their human capital is not portable and that switching banks often entails career penalties. In contrast, our results highlight a setting in which bank managers' human capital is indeed portable because it is embodied in the personal relationships they maintain with clients and continues to generate value even after they move across institutions.

Personnel economics has extensively documented the impact of managers on firm performance (Bloom et al., 2013; Bandiera et al., 2020; Lazear et al., 2015). Building on this literature, Minni (2026) shows that manager rotations can have long-term effects on subordinates' careers. We extend this perspective to the financial sector, showing that bank managers influence not only the performance of their own organization by screening and monitoring better borrowers but also that of their client firms, by facilitating access to credit and improving loan terms. The study most closely related to ours is Patault and Lenoir (2024), who defines "customer capital" for sales managers and shows that client relationships can persist when managers change employers. We extend this insight to the banking sector, where the role of soft information is especially prominent, and show that the portability of managerial relationships is stronger and more consequential. Moreover, we demonstrate that

these relationships benefit both sides of the market: banks acquire better borrowers, and firms obtain improved credit conditions.

The remainder of the paper is organized as follows. Section 2 describes the Italian institutional context, presents our data sources, and outlines the construction of the sample. Section 3 details the empirical strategy and explains how we identify the effect of manager mobility on credit allocation. Section 4 presents our main results on the portability of credit relationships, both across and within banks. In Section 5, we apply the same empirical framework to loan application data to examine whether managers have a greater impact on the application or the screening stage. Section 6 investigates how portability varies across firms, banks, and managers, highlighting the channels through which soft information passes most effectively. Section 7 analyzes how managerial mobility affects loan contract terms and loan performance, showing that managers reduce search frictions for firms, leading to better credit terms for them and lower NPL rates for banks. Finally, Section 8 concludes.

## 2 Data and sample construction

In the following paragraphs we first present the datasets we use, then we describe how we perform our matching procedure, and finally we describe the resulting dataset that we use in our empirical analysis.

### 2.1 Main datasets

We combine three different datasets, which are provided by the Bank of Italy and refer to the period 2009-2018. The first one is the credit registry data, which contains all loans granted by Italian banks to Italian firms. The second one is the social security data, which contains all the workforce of the Italian banking sector. The third one is the Cerved data, with balance sheet information on all incorporated Italian firms.

**Credit information** The credit registry data, provided by the Bank of Italy, is a panel of the loans granted by Italian banks to Italian firms in the years 2009-2018. We observe a firm's loans with a bank whenever the firm's total debt towards that bank is at least 30 thousand euros. Each loan is identified by the receiving firm, the year and the granting bank branch.[1] We observe the total amount of outstanding debt at the

[1] In this paper we identify bank branches in the credit registry by the unique combination of municipality and bank group codes.

end of each year, as well as the breakdown by type of loan (credit line, self-liquidating or term loan). We also track the status of each loan (performing or non-performing) at the end of each year. From two separate sections of the credit registry we also retrieve data on interest rates and loan applications. Interest rates are coded at the bank group - firm level, so that we can tell the aggregate average interest rate that a firm has paid to a bank group in a given year. For loan applications, we use a dataset containing the requests for information ("Richieste di prima informazione" in Italian) that banks make to the credit registry in order to screen potential borrowers or monitor existing ones (Branzoli and Fringuellotti, 2020; D'Andrea et al., 2026). We consider these requests for information as an intermediate step between the first contact between a firm and a bank, and the actual granting of a loan (Albertazzi et al., 2017; Alessandri and Bottero, 2020). The availability of such data is unique in the context of credit registry data, and allows us to investigate the mechanisms behind our main results. As for the interest rates, requests for information are coded at the bank group - firm level, so that we can tell if a bank group has requested information about a firm in a given year.

We consolidate the bank group identifier at the end of the sample, in order to avoid considering any mechanical credit relocation due to M&As.[2] The dataset contains 8 million observations (one for each firm-branch-year triplet), 440 thousand unique firms, and 25 thousand unique branches.

**Workforce information** We access a subset of Italian Social Security data, which contains the universe of workers in the Italian banking sector for the years 2009-2018. The dataset is provided to the Bank of Italy by INPS (Istituto Nazionale della Previdenza Sociale, the Italian National Institute of Social Security). We observe for each worker, at yearly level, her bank group, the municipality where she works,her type of contract (managerial or not), her wage and a set of demographic characteristics (such as age or gender). As for the credit registry data, we identify a branch by the unique combination of municipality and bank group. The dataset comprises

[2]Consider indeed the case in which Bank group A acquires Bank group B in 2015, and the two banks have a branch each in the same municipality. Right after the M&A operations the two branches would be considered as the same branch in the credit registry. Therefore we would observe credit (and also people, as we will see later) moving mechanically from one branch to the other. As we are interested only in firm's decision to move credit, we use a version of the credit registry that sets the bank group at the end of the period as the identifier of the branch for the whole period. In the previous case we then would have a unique branch of Bank group A in the municipality for the years 2009-2018.

around 3.5 million observations; restricting it to the banking sector and to the workers observed in every year of their career, it contains around 394 thousand unique workers.

**Firm characteristics** The third dataset contains firm-level characteristics for all incorporated Italian firms. It is administered by Cerved Group and licensed to the Bank of Italy. Our baseline analysis uses the full credit registry and does not rely on this dataset. We use the Cerved data only to study heterogeneity across firm characteristics, such as size, industry, and credit score. The dataset covers around 500 thousand unique firms, of which around 300 thousand also appear in the credit registry in at least one year of our sample period. Heterogeneity analyses are therefore restricted to this matched sample. The dataset provides balance sheets and income statements, which we use to construct standard firm-level variables, such as size, industry, assets, investments, credit score, and profits. Sole proprietorships, small household producers, and unincorporated partnerships are not covered. A firm is identified by a unique fiscal code, which we match to the credit registry through a crosswalk provided by the Bank of Italy.

### 2.2 Sample construction

We construct our sample by adding to the credit registry information on workforce transitions. Therefore our final dataset will be built with the same structure of the credit registry (firm-branch-year triplets), and an additional column that keeps track of manager moves, matching firms that the managers have known in the past to the branches where they move. Our matching procedure relies on the fact that we can identify where a manager is employed in the social security data, and where a firm has received credit from in the credit registry data. We measure knowledge flow through transitions, of which Figure 1 provides a schematic representation. Consider two banks, that we just label as old and new bank (with respect to the manager's move). At time $t = 0$, the manager is in her old bank, where she gives credit to firms A and B. At the same time, the new bank she will move to has credit relationships with firms C and D. At time $t = 1$, the manager moves to the new bank, and the credit relationships change. Both firm A and B are part of the manager's portfolio as she moves. Firm A actually follows the manager to the new bank, while firm B does not. In order to measure this phenomenon in the data, we need to define who the branch managers are, which firms are in their portfolio, and what a move is.

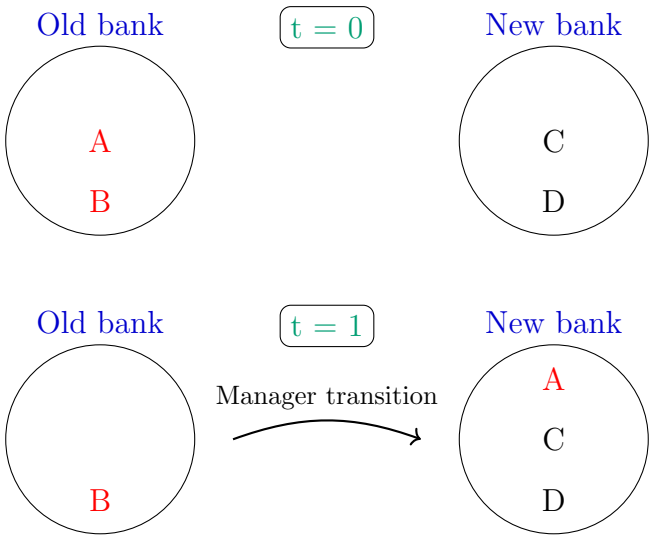



Figure 1. Representation scheme of the manager transition.

*Notes:* Figure represents the impact of a manager transition from the old bank to the new bank. Firms A and B were part of its portfolio at the old bank, A follows her.

**Branch managers** In both the credit registry and the social security data, a branch is identified by the unique combination of municipality and bank group codes. In order to define who the branch managers are, we select: the workers that are labelled as top or middle managers in the social security data, as well as those whose wage was in the top decile of the wage distribution in her branch.[3] At the same time, we need to identify within managerial workers, those who actually are in charge of their branch and make decisions about lending. As a consequence, we focus only on a particular subset of branches. First, we select only branches of bank groups that have had only one physical address in the municipality where they operate. While the credit registry (nor the social security data) does not provide physical addresses, we rely on a publicly available data source on the Bank of Italy website that provides the physical addresses of all bank branches in Italy.[4] Second, within those branches, we select the ones that had given credit to at most 150 unique firms per year in the two years before a manager moves. This restriction applies to all the analyses in the paper. The average number of managers per branch is 2.7, and we keep 80% of the branches in our sample. This choice allows us to be sufficiently sure that the manager has had direct contact with the firms which had loans with the branch, as well as

[3]Without making the assumption on wages, we would not be able to identify any manager in 35% of the branches. However, our results hold regardless of this assumption.

[4]See https://www.bancaditalia.it/compiti/vigilanza/albi-elenchi/

to avoid branches that are too large and may have multiple managers with different responsibilities. In Appendix we provide additional robustness checks to confirm that our results are robust to the threshold choice.

**Portfolio** Once we have established who the branch managers are, and what are the relevant branches, we need to pin down the portfolio of clients each branch manager has. We label as portfolio firms all the firms that have had active loans with the branch where the manager works in both the two years before the manager moves. We do it to accomplish two results: first, we want to make sure that the manager was in direct contact with those firms up until she moved (i.e. we want to exclude firms that left the branch before the manager moved). Second, we want to consider as portfolio firms only the ones who have had a sufficiently long relationship with the branch, so that the manager could have actually acquired soft information on them. The resulting average portfolio size is around 40 firms per manager. Notice that we are not able to tell which firm was interacting with which manager within the branch. Therefore, our choice of restricting to small branches is crucial, as it increases the likelihood that the manager was actually in contact with the firms that we consider as her portfolio. Consider also that the possible noise in our portfolio construction would bias our results towards zero, as we introduce in a manager's portfolio firms that she may not have known. Any measure of the portability of credit relationships that we find is therefore conservative. We test the robustness of our results by focusing on subset of branches that had a smaller number of managers, and our results are - as expected - stronger.

**Moves** The third part of our matching procedure is how to define a move. In social security data, we observe where the branch manager works in each year. We know both her municipality and the bank group she works for. Our baseline definition of a move is a change in the bank group of the manager from year $t$ to year $t+1$. Since we are interested in permanent moves and not temporary relocations, we constrain our set of moves such that at $t-1$ the manager was still in the same branch where she worked at time $t$, and at $t+2$ she is still in the branch where she moved at $t+1$. However, there is also another type of move, which is a change in the municipality of the manager within the same bank group. The two types are inherently different. In the baseline case, which concerns 7.0% of the managers who move in our data (Table 1), the manager transfers information from one bank to another, and create

a new credit access for the client firms (Gao et al., 2024b). The second case, which concerns the remaining 93.0% of the managers who move, who only change branch within the same bank group, corresponds to a more traditional definition of loan officer rotation, or may be generally due to personnel management within the bank (Minni, 2026; Gao et al., 2024a). Even in this latter case there is room for information transmission, as personal information is specific to the branch manager, and the relationship lending literature has shown that even within the same bank, loan officers have incentives to hoard information about borrowers (Hertzberg et al., 2010). Our baseline specification will consider only moves between bank groups, as they constitute a unique feature of our dataset. After the restrictions described in this section, our final sample contains around 700 moves between bank groups. We use the second type of moves in our heterogeneity analyses to (i) assess the robustness of our main findings and (ii) benchmark our results against the more traditional definition of loan officer rotation.

### 2.3 Resulting dataset

We use the definitions of branch managers and moves in the social security data, and build for every manager that has moved her portfolio of firms. Our main interest is to measure the increase in the probability that a portfolio firm gets credit from the branch where the manager has moved, after the move has taken place. In order to do so, we need to act on the credit registry to properly measure this probability. The credit registry is a panel of firm-branch-year triplets, but contains only the pairs that have realized (i.e. the firm has a loan with the branch).

We then need to modify the credit registry to account for all potential firm-branch relationships, not just the realized ones. This means we need to create a balanced panel of firm-branch-year triplets for the years 2009-2018, where each triplet represents a potential relationship between a firm and a branch in a given year.

To do this, we first filter the credit registry considering only all firms and branches that have been active (i.e. received or given credit in each year) during the period of interest. We do it to avoid any bias that may derive from the fact that some firms or branches may have entered or exited the credit market during the period of analysis. The resulting filtered credit registry contains 4 million realized matches (one for each firm-branch-year triplet), of 190 thousand unique firms and 14 thousand unique branches. All the 190 thousand firms in our filtered sample are present in Cerved data.

Then we expand this filtered credit registry to include all potential firm-branch-year triplets for the years 2009-2018. From the perspective of the firm, we need to consider all the branches it could have asked credit from, consider the ones that have realized, and measure if the presence of a manager that knew the firm has affected this matching realization. Similarly, from the perspective of the branch, we need to consider all the possible firms it could have given credit to. Panetta et al. (2009), Crawford et al. (2018) and Barone et al. (2024) define credit market at the provincial level,[5] so that we consider only firm-branch potential pairs that are in the same province. Then our final dataset is the aggregation of the 110 provincial markets, where a market is defined as the set of all potential firm-branch pairs in a province. Each potential firm-branch pair is represented in each year from 2009 to 2018.

The dataset has a size of 231 million observations, which makes computationally expensive the estimation of our model. In our main regressions we sample around 40% of the data, and we repeat our exercises to validate our results.[6]

Table 1 describes our data. The first column refers to the full data: about 442 thousand firms in the credit registry, which borrow on average from 2.3 bank groups; almost 15 thousand branches of 436 bank groups; and about 394 thousand workers in the banking sector, of whom 183 thousand are managers in at least one year. The second column restricts to the qualifying branches, i.e. the branch-years in which the bank has a single physical branch in the municipality and fewer than 150 client firms, which are the branches whose managers can enter our analysis, and to their client firms and workers. They account for about 70% of all branches. Their clients are larger borrowers than the average firm (a median credit of 465 thousand euros, against 311 thousand) and have relationships with more bank groups (almost 3 on average), while the branches themselves are smaller, with 27 client firms and fewer than 3 managers per year on average.

The last panel of Table 1 documents the moves of managers over the period 2009-2018. Out of about 183 thousand managers, 75 thousand (41%) change branch at least once. Of these, 5,287 (7.0%) move at least once between bank groups, while the remaining 69,728 (93.0%) only move within the same bank group. Among the managers who work in, and move out of, qualifying branches, 640 move between bank groups. The remaining restrictions (a move within the province, a stable stay in the

[5] See the discussion in the previous paragraph.

[6] We set a seed for reproducibility. We keep all treated firms and randomly sample 10% of control firms, retaining all branch-firm pairs of each sampled firm. This procedure results in a 40% sampling coverage.

branch before and after the move, a destination bank with one physical branch in the municipality, and a portfolio of always-active firms with fewer than 150 firms) leave the 680 moves of our final sample.

Table 1. Descriptive statistics

| | Full data | Qualifying branches |
|---|---|---|
| *Firms* | | |
| Number of firms | 442,186 | 271,228 |
| Credit per firm-year (thousand euros), mean | 2,271 | 3,178 |
| median | 311 | 465 |
| Credit relationships per firm-year, mean | 2.34 | 2.95 |
| *Banks and branches* | | |
| Number of bank groups | 436 | 381 |
| Number of branches | 14,702 | 10,149 |
| Credit per branch-year (thousand euros), mean | 40,082 | 25,408 |
| Firms per branch-year, mean | 43.6 | 27.3 |
| *Workers* | | |
| Total number | 394,389 | 149,788 |
| managers | 182,619 | 65,765 |
| Managers per branch-year, mean | 8.5 | 2.7 |
| *Moves* | | |
| Managers with at least one move | 75,015 (41.1%) | 15,342 (23.3%) |
| at least one between-group move | 5,287 (7.0%) | 640 (5.4%) |
| only within-group moves | 69,728 (93.0%) | 14,662 (94.6%) |

*Notes:* Totals and means over 2009-2018. Full data: all firms in the credit registry, all branches (a branch is a bank group × municipality cell) and all workers in the banking sector observed in every year of their career. Qualifying branches: branch-years in which the bank has one physical branch in the municipality and fewer than 150 client firms (always-active firms with an original or mature credit relationship); firms are the clients of a qualifying branch in that year (their statistics use all their credit), branch statistics refer to the bank group × municipality cells containing a qualifying branch, workers are those employed in a qualifying branch, and moves are moves out of a qualifying branch. Credit is the amount granted (accordato); credit relationships are counted at the bank group level. Managers follow the definition in Section 2.2. A move is a change of branch between two consecutive years by a worker who was a manager in the year before: within bank group if the bank group (defined at the end of the sample) is the same and the municipality changes, between bank groups if the bank group changes. Percentages in parentheses: managers with at least one move, as a share of managers; managers with at least one between-group move and managers with only within-group moves, as a share of managers with at least one move.

## 3 Empirical Strategy

Our main goal is to quantify how portable are the relationships that bank managers build with their clients. We measure it by computing the probability of a firm establishing a credit relationship with the new bank of the manager, as opposed to any other bank in the same province. As explained in Section 2, in order to measure this probability, we have to create matching pairs of firms and banks. With respect to banks, we can identify the branch through the unique combination of municipality and bank group. We denote a branch by $b$ and, in the analyses where data are aggregated at the bank group level, we denote the bank group to which branch $b$ belongs by $g(b)$. Then our unit of analysis is a potential match between a branch and a firm. It is treated when the branch hires a manager that has given credit to the firm in the past (i.e. the firm is in the manager's portfolio).[7] Concretely, our data form a panel at the firm-branch-year level: for each firm and year, we include all branches located in the firm's province in that year, and, for each firm-branch-year, we record (i) whether the firm has credit with the branch, and (ii) whether the branch has hired, in that year or earlier, a new manager whose previous portfolio included the firm. We rely on two sources of variation: the cross-section (i.e. movements from one branch to another) and the timing of the moves. We employ a difference-in-differences strategy, explained in Equation (1).

$$I(\text{credit})_{bft} = \beta I(\text{Inflow})_{bft} + \alpha_{bf} + \delta_{bt} + \gamma_{ft} + \epsilon_{bft} \tag{1}$$

$I(\text{credit})_{bft}$ is an indicator equal to one if firm $f$ has credit granted by branch $b$ in year $t$ (i.e. an active credit relationship), and zero otherwise. $I(\text{Inflow})_{bft}$ is an indicator equal to one if the branch $b$ has hired a manager that had firm $f$ in her portfolio. It is equal to one for all the years after the pair has been treated for the first time (i.e. for the first time a manager knowing firm $f$ has moved to branch $b$). Notice that since we are using worker relocations, it can be possible that a pair is treated more than once, as two or more managers having the same firm in portfolio can move to the same branch, contemporaneously or in different years. Notice also that the timing of the event is specific to the firm-branch pair, i.e. some units are treated at the beginning of the sampling period, and some in later years.

We include a vast array of fixed effects, that control for the time-invariant characteristics of the firm-branch relationship, and for the time-varying policies of the

[7] See the discussion in subsection 2.2 for the definition of the portfolio.

branch and the firm. More extensive discussion on the identification strategy and the use of fixed effects is provided in subsection 3.1. We double-cluster standard errors at the bank group and firm level. Treated pairs are those formed by the manager's new branch and each firm in her portfolio, from the year of the move onwards, whether or not the firm actually follows her. All other firm-branch pairs in the same province serve as controls, including those formed by portfolio firms with all other branches.[8]

The coefficient $\beta$ measures the increase in the probability of a firm establishing a credit relationship with the new branch of the manager, as opposed to any other branch in the same province. In our main specification we will only rely on variation coming from moves between bank groups, that have a more profound impact on information transmission throughout the financial system. However, we will use the same empirical strategy to detect the effect of moves within the same bank group, both to consolidate and benchmark our results, and to inform the literature on loan officer rotation.[9]

We estimate Equation (1) also through an event study specification. It allows us to provide a visual representation of our main results, and to rule out any anticipation of the move by the firms. If indeed the branch where the manager was working had poor performance, firms may have reacted to such poor performance by moving their credit relationship elsewhere. And if the manager was aware of the poor performance, she may have decided to relocate exactly to the branch where the firm had already moved.

Our setup allows us to test for this anticipation effect, in two ways. First, if the firm had relocated by shutting down the credit relationship with the manager's old branch before her move, it would not be part of the portfolio, and hence not treated. Second, if the firm had relocated but kept the credit relationship with the manager's old branch at least until the time of the move, we would observe a higher probability of the firm establishing a credit relationship with the new branch before the event. This would be a sign of anticipation of the move, and would suggest that the manager's move was not the cause of the firm's relocation. Equation (2) provides a formal representation of the event study specification.

[8] For more details on the sample construction, see subsection 2.3.

[9] Pairs treated by moves that are not part of the estimation sample (for instance, within-group moves when we estimate the effect of moves between bank groups, and vice versa) are part of the control group. Since within-group moves also increase the matching probability (Section 4.2), this choice, if anything, biases our estimates towards zero.

$$I(\text{credit})_{bft} = \sum_{\tau=-4}^{4} \beta_\tau \times I\{t = t_{bf} + \tau\} + \alpha_{bf} + \delta_{bt} + \gamma_{ft} + \epsilon_{bft} \quad (2)$$

The term $t_{bf}$ is the year in which the manager that knows firm $f$ moves to branch $b$. $I\{t = t_{bf} + \tau\}$ is an indicator equal to one if the year is $\tau$ years before or after the manager's move. The coefficient $\beta_\tau$ measures the year-specific variations in the matching probability between the firm and the branch. We standardize $\beta_{-1}$ to zero, so that the other coefficients represent variation with respect to the baseline year. All fixed effects, sample and control group are the same as in Equation (1). We take into account the staggered nature of the treatment, using estimators that are robust to negative weights or cohort-specific (heterogeneous treatment) effects (Sun and Abraham, 2021).

### 3.1 Identification discussion

We exploit the dyadic structure of our data, and include three types of fixed effects.

First, we include pair fixed effects $\alpha_{bf}$, that control for the time-invariant characteristics of the relationship between the firm and the branch. In the context of our analysis, they represent the assortative matching probability between firms and branches. They control for geographical characteristics - i.e. firms that are close to the branch are more likely to have a loan with the branch, as in the relationship lending literature (Bonfim et al., 2021; Keil and Ongena, 2024; Nguyen, 2019) - or industry components - in the case of lending specialization, as in Paravisini et al. (2023); Duquerroy et al. (2022); Goedde-Menke and Ingermann (2023).

Second, we include branch-time fixed effects $\delta_{bt}$, that take into account for branch-level time-varying policies, such as credit supply, branch size (hirings or layoffs), deposit inflows or outflows.

Third, as Khwaja and Mian (2008), we include firm-time fixed effects $\gamma_{ft}$, which account for firm-level time-varying characteristics, such as credit demand, investments, or financial health of the firm. A unique feature of the Italian banking sector is the presence of multi-bank relationships, even for small firms (the average number of bank relationships per firm in the credit registry is 2.76), as documented also by Gobbi and Sette (2014) and Kosekova et al. (2024). Therefore, the inclusion of firm-time fixed effects appears to be even more crucial in this context, as firms may direct their credit demand to different banks and branches over time. Differently from Khwaja and Mian (2008), who study loans within existing relationships, our outcome

is whether the firm has a credit relationship with a given branch, which for pairs with no prior relationship captures its formation. Firm-time fixed effects absorb firm-level demand but not demand directed at a specific branch, so $\beta$ captures match-specific variation, which Section 5 separates into applications and approvals.

Then the parameter $\beta$ is identified by any variation that, at the event time, changes the matching probability between the firm and the branch.

Absent any confounders, we attribute the change in the matching probability to the manager's move, as Patault and Lenoir (2024). However, there is still a threat to our identification strategy, coming from the potential endogeneity of the manager's move. Indeed, the decision of the manager to move may be driven by shocks that also change the matching probability between the firm and the branch. An example of that is some poaching activity, where the new bank actively tries to attract the firms of the manager (and maybe wants to expand in the same industry), and as a consequence the manager decides to move. An event study regression can help us control if the managers' new branch was already attracting the firms in the portfolio before the move, but cannot rule out simultaneity. $\beta$ would not correctly identify the portability of the relationship if, at the same time, but separately, both the manager and the firm decided to move.

In this setting, we address this concern by estimating equation (2) by restricting our sample of moves to those that are caused by branch closures, as in Eliason et al. (2023). Italian labor law tries to protect workers from the consequences of branch closures, and facilitates their relocation within the financial sector. Most of the relocations happen within the same bank group, but we discard them for two reasons. First, in our main specification we are interested in bank-to-bank moves. Second, when a bank group closes a branch, its borrowers are automatically transferred to another branch of the same bank group (Papoutsi, 2025).

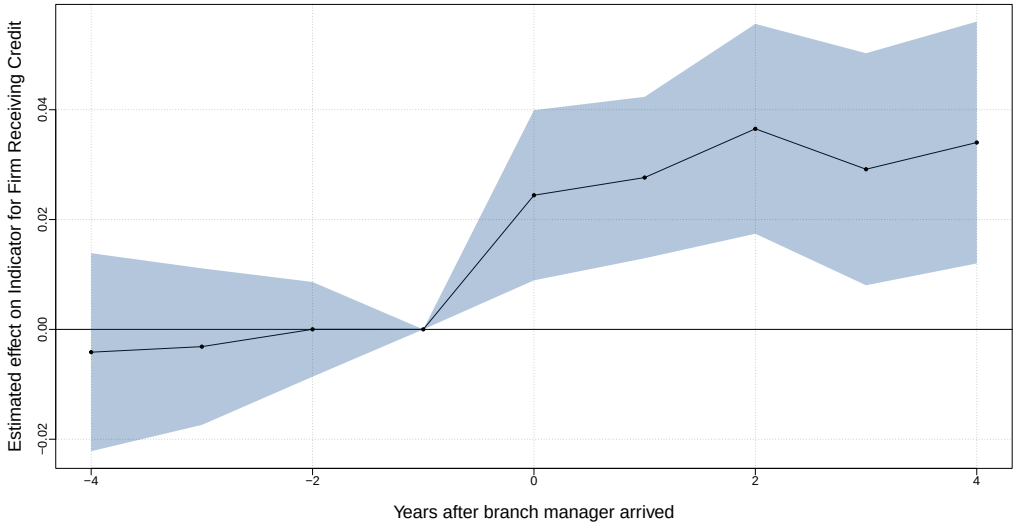
We want to avoid any mechanical credit relocation, so we consider only moves in which the manager joins a different bank group. Since bank groups are defined at the end of the sample, these automatic transfers never involve the manager's new branch. By including fixed effects, and restricting to moves triggered by branch closures, we are thus able to identify the effect of the manager's knowledge on the matching probability between the firm and the branch.

# 4 Main Results

Figure 2 shows the event study representation for Equation (2). It represents the time variation of the probability that the branch $b$ that hires the manager establishes a credit relationship with the firm $f$ that was in the manager's portfolio. As one can notice, there is no significant variation in the probability of creating such relationship before the manager moves. To provide a benchmark for our estimates, consider that the average probability of a firm establishing a credit relationship with any branch in the same province is around 1.3%. In the year right after the manager moves, the probability increases by 2.4 percentage points, and then by more than another percentage point in the following years. One can notice two main facts. First, the increase is sizable: the probability almost triples relative to the baseline, from 1.3% to about 3.7%. Second, the dynamics show that managers are able to bring their clients with them right after they move, as most of the increase happens in the first two years.[10] Put differently, four years after the move the probability that a firm in the manager's portfolio has a credit relationship with her new branch is around 5%, almost four times the baseline of 1.3% (a 3.7 percentage point increase), so that, out of an average portfolio of 40 firms, about two firms have a relationship with the manager's new branch.

[10]Notice that obtaining credit in Italy is rather a fast process, with the average time between the application and the disbursement of the loan being around 30 days.

Figure 2. Increase in probability of giving credit to portfolio firms after the move




*Notes:* Event study estimates for the probability of a firm receiving credit from the branch where its previous loan officer has moved. Plot shows OLS estimates and 95% confidence intervals for the coefficients $\beta_\tau$ in equation (2). Branch - firm, branch - time, and firm - time fixed effects are included. Standard errors are clustered at the bank group - firm level. Sample includes all potential firm - branch pairs within a province, for the period 2009-2018. Only firms and branches that are present in the data in all years are included. Moves are defined as relocations of managers from one bank group to another. To keep consistency with the sample definition, only moves that occur within the same province are considered. Estimates are obtained using the Sun and Abraham (2021) estimator.

Table 2 shows the DiD estimates for $\beta$ across different specifications. The first column shows the baseline specification, with just the inclusion of branch-firm fixed effects, to capture the assortative matching time invariant characteristics. In columns (2) and (3) we include branch-time and firm-time fixed effects, while column (4) shows the results with the whole set of fixed effects. Estimates are stable across all specifications, indicating that our results are not driven by credit demand or supply shocks. In particular, the estimate is, if anything, slightly larger in column (3), which does not compare portfolio firms with other firms at the same branch, so our results are not driven by the new branch displacing its existing clients. The coefficient hovers around 0.025.

Table 2. DiD estimates for $\beta$

| | Credit indicator | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Inflow | 0.027*** | 0.022*** | 0.026*** | 0.023*** |
| | (0.007) | (0.007) | (0.007) | (0.007) |
| $R^2$ | 0.772 | 0.785 | 0.773 | 0.786 |
| Observations | 44,681,890 | 44,681,890 | 44,681,890 | 44,681,890 |
| Dependent variable mean | 0.013 | 0.013 | 0.013 | 0.013 |
| Branch-Firm fixed effects | ✓ | ✓ | ✓ | ✓ |
| Branch-Time fixed effects | | ✓ | | ✓ |
| Firm-Time fixed effects | | | ✓ | ✓ |

*Notes:* Estimates for the probability of a firm receiving credit from the branch where its previous loan officer has moved. Column (1) includes only branch fixed effects, column (2) includes branch and time fixed effects, column (3) includes branch and firm fixed effects, and column (4) includes all fixed effects. Standard errors are clustered at the bank group - firm level. The sample includes all potential firm - branch pairs within a province, for the period 2009-2018. Only firms and branches that are present in the data in all years are included. Moves are defined as relocations of managers from one bank group to another. To keep consistency with the sample definition, only moves that occur within the same province are considered. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

### 4.1 Branch closures identification

In order to address the potential endogeneity of the manager's move, we estimate Equations (1) and (2) considering only moves that are caused by branch closures. As in the previous case, we consider only bank-to-bank moves. It is noteworthy that considering only bank-to-bank moves allows us to rule out any form of mechanical credit relocation. Indeed, if a bank group decides to close a branch, it will move both the managers and the credit relationships to a different branch. By considering only managers that have moved to a different bank group, we rule out automatic transfers of credit relationships within the bank group.

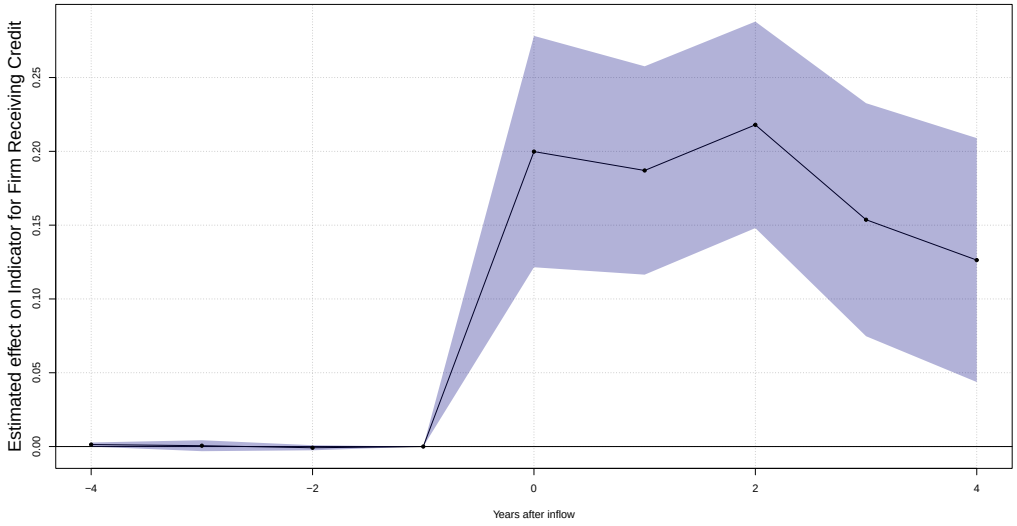
Using branch-closure induced moves as a source of identification helps us to address the potential endogeneity of the manager's move. However, we expect a different

dynamic pattern. We can decompose the probability of a firm following the manager into two components: the probability that the firm decides to leave its current branch, and the probability that the firm decides to apply to the manager's new branch.

In the case of branch closures, the first component is likely to be higher, as the firm is more likely to search for credit elsewhere. Firms are automatically relocated to a different branch of the same bank group (Papoutsi, 2025), so they can decide whether to accept this relocation or to search for credit elsewhere. In the latter case, they may decide to follow the manager to her new bank or to simply shop around. Therefore, we expect that the overall effect of the manager's move on the matching probability is higher, and more sudden, than in the case of all bank moves.

Figure 3 shows the event study representation. As for the previous case, there is no anticipation of the move by the firms, that do not move to the new branch before the manager has moved. The right-hand side of the graph confirms our hypotesis: the matching probability increases more rapidly than the case of all bank moves, then stays stable for two years, and slightly decreases in years 3 and 4. The overall effect is much larger than the previous case, hovering around an increase by 20 percentage points in the matching probability. We attribute this larger magnitude exactly to the augmented separation probability from the previous branch, that forces firms to search for credit elsewhere. The decrease in years 3 and 4, although present, still leaves the matching probability 10 percentage points higher than the baseline probability. Appendix Table A2 reports the DiD estimates with the progressive inclusion of fixed effects, that - as in the previous case - are stable across specifications.

Figure 3. Increase in credit probability to portfolio firms after branch closures




*Notes:* Event study estimates for the probability of a firm receiving credit from the branch where its previous loan officer has moved due to a branch closure. Plot shows OLS estimates and 95% confidence intervals for the coefficients $\beta_\tau$ in equation (2). Branch - firm, branch - time, and firm - time fixed effects are included. Standard errors are clustered at the bank group - firm level. Sample includes all potential firm - branch pairs within a province, for the period 2009-2018. Only firms and branches that are present in the data in all years are included. Moves are defined as relocations of managers from one bank group to another, and the originating branch has closed. To keep consistency with the sample definition, only moves that occur within the same province are considered. Estimates are obtained using the Sun and Abraham (2021) estimator.

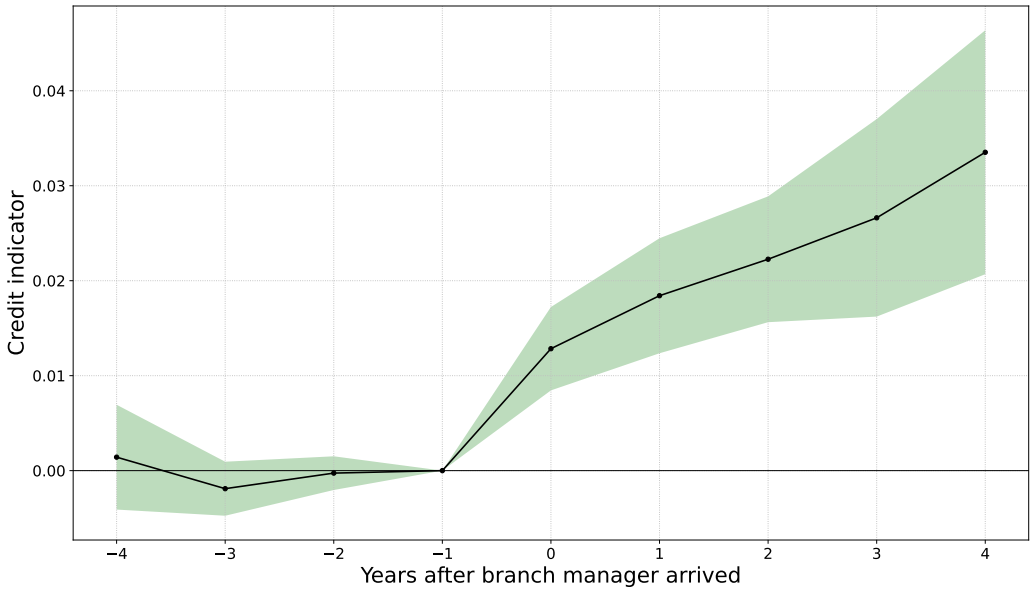
### 4.2 Within bank moves

We now turn to the case of moves within the same bank group. Moves between bank groups are most likely caused by career choices of the manager, who actively sought a new position, or by external circumstances such as branch closures. Here we consider a different type of transition: within group moves are often decided by the bank. Managers can be rotated across branches, or promoted to a different position, maybe in a larger or more strategic branch. The case of loan officer rotation has been vastly studied by theoretical (Stein, 2002) and empirical literature (Hertzberg et al., 2010), and is a policy designed by the bank to ensure a more trasparent monitoring, reducing the incentives for loan officers to grant credit to undeserving borrowers. However, notice that the presence of personal ties between the loan officer and the borrower can also have a positive impact on the bank's profits. Indeed, more profound knowledge between the two parts may increase the probability that the borrower stays

with the bank, and can even increase the loan performance outcomes (Fisman et al., 2017). So, while an effective design of such policies would rely on the fact that the loan officer is not able to bring her clients with her, it is not straightforward that the bank would reduce this possibility to zero.

In Figure 4 we show, indeed, that even in the case of within group moves, the manager is able to bring some of her clients with her. Table A3 in the Appendix shows the DiD estimates for the within group moves. The estimated effect is of a magnitude comparable to the case of between group moves (Table 2), but it exhibits a different dynamic pattern: in the case of between group moves, the welcoming bank has no incentive to delay the transfer of new clients (if valuable). Also in this case firms do not anticipate the move. But the matching probability, instead of increasing sharply right after the move, increases constantly, and peaks five years after, with an increase of more than three times the baseline probability. We can interpret it as the interplay of two forces: while the manager may like to bring her clients on, the bank may have policies that prevent internal relocations. So, for the manager it may take a longer time to progressively bring her clients with her.

Figure 4. Credit probability increases, within bank moves




*Notes:* Event study estimates for the probability of a firm receiving credit from the branch where its previous loan officer has moved. Only moves within the same bank group are considered. Plot shows OLS estimates and 95% confidence intervals for the coefficients $\beta_\tau$ in equation (2). Branch - firm, branch - time, and firm - time fixed effects are included. Standard errors are clustered at the bank group - firm level. Sample includes all potential firm - branch pairs within a province, for the period 2009-2018. Only firms and branches that are present in the data in all years are included. Moves are defined as relocations of managers within the same bank group. To keep consistency with the sample definition, only moves that occur within the same province are considered.

This result provides a benchmark for the previous estimates: the DiD estimates for within bank moves (Table A3) are close to those for bank moves (Table 2), so the magnitudes are comparable, while the dynamics differ. Between bank portability builds up right after the move, whereas within bank portability takes longer to materialize. We can at least partially attribute this difference in dynamics to the bank's incentives. It also complements the existing literature on loan officer rotation. In the period covered by our sample, the number of bank groups has decreased. In our data we set the bank group at the end of the period, so that our results are net of any merger and acquisition activity. As a consequence, some of the within bank moves that we observe have happened in the context of consolidation processes. In such events, banks may have an incentive to exploit their managerial human capital to retain clients (which is what we observe in our results), but at the same time this fact reduces the monitoring effectiveness (Hertzberg et al., 2010).

In the following section we focus on the process by which the bank that hires the

manager decides to grant credit to the firms that were in the manager's portfolio. We use data from loan applications to see if the manager is able to affect the screening process of the bank, and if the bank is able to capture the value of the manager's portfolio.

## 5 Mechanism: credit applications

In the previous section we provided evidence on the fact that when a manager moves, the firms that were in her portfolio are more likely to establish a credit relationship with her new branch. In this section we uncover the mechanism behind this phenomenon. We still focus on bank-to-bank moves, and we rely on a different data source from the credit registry, which is a request for information that the bank sends to the Bank of Italy. The object of the request is a firm, and the Bank of Italy provides the requesting bank with the firm's credit history, or its non-performing loans. It is a tool that can be used both in screening a potential borrower, so to acquire more information on the firm, or in monitoring the firm's behavior in the following years (Branzoli and Fringuellotti, 2020; D'Andrea et al., 2026). We focus on requests for information that were filed from banks to firms that did not have previous credit relationships with the bank, so to isolate the screening component of the request. This differs from Branzoli and Fringuellotti (2020), who use requests on existing borrowers to measure monitoring. Noting that filing request for information is a rather standard procedure in the screening process, we can also interpret those requests as a proxy for loan applications, as in Albertazzi et al. (2017) and Alessandri and Bottero (2020). Indeed, since requests for information come from the bank, we can reasonably assume that they come after the screened firm has previously applied for a loan.

The Bank of Italy stores requests for information data at the firm - bank group - year level. We are not able to identify the branch from which the request comes, but we can still reliably assume that the request is sent by bank where the manager has moved, and where the firm will eventually have credit, if the request is accepted. Notice also that we are interested in differences in the probability of the request being sent (or accepted), so measurement error is constant throughout the sample, and does not affect our estimates. We make two conjectures on the manager's role in the screening process. First, we expect branch managers to influence search, i.e. to induce the firms that were in her portfolio to apply for a loan. Second, we expect managers to influence the screening process, i.e. to increase the probability of the

request being accepted.

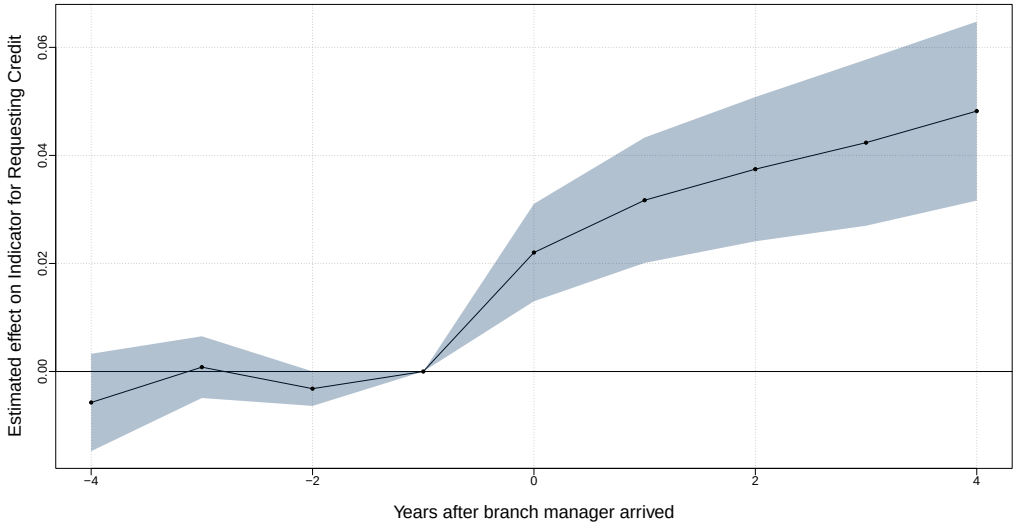
### 5.1 Search

Proxing loans applications with requests for information, we can test if the manager is able to affect the probability of the firm applying for a loan. If this is the case, we should see an increase in the requests of portfolio firms after the manager moves. Equation (3) shows the empirical strategy we use to test this conjecture.

$$I(\text{applied})_{g(b)ft} = \sum_{\tau=-4}^{4} \beta_\tau \times I\{t = t_{g(b)f} + \tau\} + \alpha_{g(b)f} + \delta_{g(b)t} + \gamma_{ft} + \epsilon_{g(b)ft} \qquad (3)$$

The dependent variable $I(\text{applied})_{g(b)ft}$ is an indicator equal to one if the bank group $g(b)$ has requested information on firm $f$ in year $t$ or earlier, and zero otherwise. The right-hand side of the equation is the same as in Equation (2), with the only difference that observations are collapsed at the firm-bank group level, so that the bank dimension is indexed by $g(b)$, the bank group of branch $b$. The unit of observation becomes then the bank group-firm pair, as opposed to the branch-firm pair in the previous analysis. Therefore fixed effects capture the underlying matching probability between a bank group and a firm, and the time-varying characteristics of the bank group (as well as the firm). Figure 5 shows the event study representation of the results. The baseline probability of a bank requesting information on a firm is 2.5%. Four years after the manager moves, the probability increases by 4.5 percentage points, reaching a level that is almost three times the baseline probability.

Figure 5. Increase in probability of applying for credit to the new bank




*Notes:* Event study estimates for the probability of a firm applying for credit in the bank where its bank manager has moved. Plot shows OLS estimates and 95% confidence intervals for the coefficients $\beta_\tau$ in equation (3). Bank group - firm, bank group - time, and firm - time fixed effects are included. Standard errors are clustered at the bank group - firm level. Sample includes all potential firm - bank pairs, for the period 2009-2018. A pair is considered potential if in the province where the firm was present, there was at least one branch of the bank. Only firms and branches that are present in the data in all years are included. Moves are defined as relocations of managers from one bank group to another. To keep consistency with the sample definition, only moves that occur within the same province are considered. Estimates are obtained using the Sun and Abraham (2021) estimator.

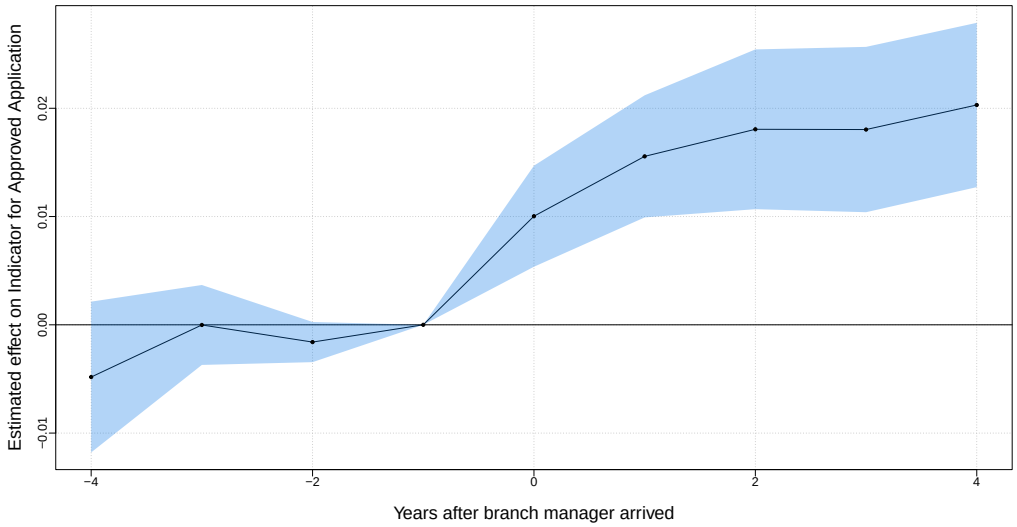
### 5.2 Screening

Our second conjecture is that the manager not only affects the probability of the firm applying for a loan, but also the probability of the request being accepted. In order to test this conjecture, we estimate Equation (4), by regressing the conditional probability of the request being accepted on the manager's move. We build the conditional acceptance probability $I(\text{accepted}|\text{applied})_{g(b)ft}$ in our dataset by checking the rows in our dataset where the firm has requested for information, and the bank group has granted the loan. The equation reads as follows:

$$I(\text{accepted}|\text{applied})_{g(b)ft} = \sum_{\tau=-4}^{4} \beta_\tau \times I\{t = t_{g(b)f} + \tau\} + \alpha_{g(b)f} + \delta_{g(b)t} + \gamma_{ft} + \epsilon_{g(b)ft} \quad (4)$$

The conditional acceptance probability mean in our sample is 0.35, which means that the bank accepts a loan request in 35% of the cases. Figure 6 shows the event study plot. Four years after the manager has moved, the probability of the request being accepted increases by 2 percentage points, reaching a level that is almost 6% higher than the baseline probability. Therefore, not only the manager's firms are more likely to apply for a loan, but also the bank is more likely to accept the request.

Figure 6. Increase in probability of being approved for credit in the new bank




*Notes:* Event study estimates for the probability of a firm being approved for credit in the bank where its bank manager has moved. Plot shows OLS estimates and 95% confidence intervals for the coefficients $\beta_\tau$ in equation (4). Bank group - firm, bank group - time, and firm - time fixed effects are included. Standard errors are clustered at the bank group - firm level. Sample includes all potential firm - bank pairs, for the period 2009-2018. A pair is considered potential if in the province where the firm was present, there was at least one branch of the bank. Only firms and branches that are present in the data in all years are included. Moves are defined as relocations of managers from one bank group to another. To keep consistency with the sample definition, only moves that occur within the same province are considered. Estimates are obtained using the Sun and Abraham (2021) estimator.

Appendix figures A3 and A4 show the results of the same exercise using a Poisson pseudo-maximum likelihood model instead of a linear probability model. The estimates confirm the main results: pre-trends are flat, and the effects are positive and significant after the manager's arrival.

In the following sections we will explore the features of the match between the manager and the firm, and will investigate what kind of value the manager brings to the bank that hires her.

# 6 Heterogeneity

We provide evidence on the three actors that are involved in the relationship lending process: the bank manager, the bank, and the firm. We uncover their role in the portability of the manager's clients by interacting the treatment effect in equation (1) with characteristics of the three actors. Throughout this section, the baseline effect is the difference-in-differences estimate of $\beta$ in Table 2 (2.3 percentage points), which averages the effect over all the years after the move and is therefore smaller than the four-year effect of 3.7 percentage points discussed in Section 4.

First, we look at the characteristics of the bank, and we show that the structure of information within both the welcoming and the departing bank matters for the portability of the manager's clients. Second, we look at the characteristics of the firms, and we show that younger and smaller firms are more likely to follow their manager when she moves, being the actors for which the manager's soft information is more valuable (Huber, 2021). Third, we look at managers, showing that older managers coming from smaller branches bring a larger fraction of their clients with them. We also show that former loan size matters, with managers bringing a larger fraction of their clients that had mid-size loans (between 100 and 500 thousand euros). With respect to competition, we show that the portability of the manager's clients is largest in the most competitive local banking markets. Taken together, these dimensions offer several pieces of evidence that are consistent with managers transferring soft information when they move.

## 6.1 Bank organization

The first dimension of heterogeneity we investigate on relates to the characteristics of the banks. As Stein (2002) explains, the structure of information within a firm has a crucial role on the internal organization. He argues that his model is mostly suited for the banking sector, where there is the contemporaneous presence of hard and soft information. Some banks - the large ones in the model - may rely more on hard information, and so have a centralized credit awarding process, where the role of the branch is limited to the collection and the "hardening" of credit information, so that it can be used by the central office. On the other hand, another type of banks - typically smaller - may rely more on soft information, and so have a decentralized credit awarding process, where the branch manager has a more prominent role in the credit allocation.

We test this conjecture by interacting the manager's move with a dummy that captures the size of the bank. We define large banks as banks that have more than 5 percent of the total number of employees in the banking sector, and small banks as the remaining ones. Since all our analyses only use branches that lend to at most 150 firms (Section 2.2), large banks are represented here by their small branches. There are then 4 categories, interacting the previous bank size (large or small) with the size of the manager's new bank.

In this context, Stein (2002) would predict that small banks are more likely to let the manager bring her clients with her, as they rely more on soft information. At the same time, they are the ones that are more willing to keep the clients within the bank, as they may implement more devices to keep the information within the bank, such as having the client firm interact with more than one manager, so to build a relationship that is personal (i.e. with the manager) but ties the client to the bank as well. Therefore, we expect that transitions that attract the most clients are the ones from large to small banks, as the information is "softened" by the moving manager and the new bank is willing to use it. On the other hand, transitions from small to large banks are expected to be the least effective, as the manager would be less able to use her soft information in the new bank, and the new bank would be less willing to let the manager bring her clients with her. Table 3 shows the results. For each regression we estimate the baseline effect (first row of the table), and then the interaction term between the manager's move and the bank size.

Consistently with the theory, we find the strongest effect for movements from a large bank to a small bank. If moving from a large to a small bank, manager's clients are 4.2 percentage points more likely to follow her than after other types of moves, which is around double the baseline effect of 2.3 percentage points in Table 2. The effect is statistically significant at the 10% level. The other estimates are noisier, but smaller (or even negative) in magnitude, reflecting the fact that the process of "softening" creditor information (as for movements from large to small banks) is more effective than the opposite "hardening" process (as for movements from small to large banks). Interestingly, the estimates for transitions between large banks exhibit a positive (although not significant) interaction term, hinting that the relevant dimension of heterogeneity may be the size of the leaving bank, rather than the size of the welcoming bank.

Table 3. Heterogeneity by type of bank move

| | Credit indicator | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Inflow | 0.013*** | 0.023*** | 0.026*** | 0.039** |
| | (0.004) | (0.008) | (0.008) | (0.016) |
| Big to small × Inflow | 0.042* | | | |
| | (0.022) | | | |
| Big to big × Inflow | | 0.021 | | |
| | | (0.017) | | |
| Small to big × Inflow | | | -0.029 | |
| | | | (0.022) | |
| Small to small × Inflow | | | | -0.026* |
| | | | | (0.015) |
| $R^2$ | 0.793 | 0.793 | 0.793 | 0.793 |
| Observations | 97,198,970 | 97,198,970 | 97,198,970 | 97,198,970 |
| Firm-Time fixed effects | ✓ | ✓ | ✓ | ✓ |
| Branch-Time fixed effects | ✓ | ✓ | ✓ | ✓ |
| Branch-Firm fixed effects | ✓ | ✓ | ✓ | ✓ |

*Notes:* DiD estimates for $\beta$ in equation 1 for different types of bank moves. Each interaction term is relative to the effect of all other types of moves. Column (1) includes an interaction term with moves from a large bank to a small bank, column (2) includes an interaction term with moves from a large bank to a large bank, column (3) includes an interaction term with moves from a small bank to a large bank, and column (4) includes an interaction term with moves from a small bank to a small bank. Small banks are defined as the ones which have less than 5 percent of the total number of employees in the banking sector. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Each interaction term in Table 3 is measured relative to the average effect of the other three types of moves, so the coefficients are not directly comparable across columns. Adding each interaction to the corresponding baseline coefficient gives the effect for each type of move: roughly 0.055 for moves from large to small banks, 0.044 from large to large, close to zero from small to large, and 0.013 from small to small. The negative interaction for small to small moves therefore reflects an effect that is smaller than average, not a negative one. Portability is high when the manager leaves a large bank and low when she leaves a small bank, whatever the destination.

Two forces are consistent with this pattern. First, firms that already borrow from a small bank already benefit from relationship lending, so the manager's move adds less. Second, as discussed above, small banks implement devices to keep information within the organization, such as having several managers interact with the same client, which limit what the departing manager can take with her. In large banks, which rely more on hard information, the soft information held by the manager is harder to use, and their clients may be credit constrained relative to their true quality. A move to a small bank that can use this information relaxes that constraint. This is consistent with the stronger effects for younger and smaller firms that we document in Section 6.2.

We also analyse the role of the internal organization of the bank. Independently of the size of the bank, local headquarters may employ managers that perform a wider range of tasks, and so they may be less likely to bring their clients with them. If soft credit information was easily "hardenable", then managers moving to local headquarters would be as effective as managers moving to smaller branches. On the other hand, if soft credit information still resides in individuals, that have to perform credit allocation tasks even in local headquarters, then managers moving to local headquarters would be less effective. We use branches located in the capital of the province (capoluogo di provincia) as a proxy for local headquarters. There are 110 provinces in Italy, and so 110 provincial capitals, and provinces correspond to a standard definition of local credit market in Italy (Crawford et al., 2018). We then interact the manager's move with a dummy that captures if the manager moved to a branch in a provincial capital. Table 4 shows the results.

Table 4. Heterogeneity by moves to provincial capitals

| | Credit indicator | |
|---|---|---|
| | (1) | (2) |
| Inflow | 0.044*** | |
| | (0.016) | |
| Capoluogo × Inflow | -0.035** | 0.009* |
| | (0.016) | (0.004) |
| $R^2$ | 0.786 | 0.786 |
| Observations | 44,681,890 | 44,681,890 |
| Dependent variable mean | 0.013 | 0.013 |
| Firm-Time fixed effects | ✓ | ✓ |
| Branch-Time fixed effects | ✓ | ✓ |
| Branch-Firm fixed effects | ✓ | ✓ |

*Notes:* DiD estimates for $\beta$ in equation 1 for different types of bank moves. Column (1) includes an interaction term with moves to provincial capitals, and column (2) contains only the interaction term. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The estimate for the interaction term is negative and significant, suggesting that managers moving to provincial capitals are less likely to bring their clients with them. Notice that also the magnitude is relevant: as shown in column (2), that contains only the interaction term, managers moving to provincial capitals bring less than 1 percent of their clients with them, which is around one third of the baseline effect of 2.3 percentage points in Table 2.

### 6.2 Firm characteristics

We then turn to the second actor in the relationship lending process: the firm. We examine three aspects of the firms: size, age, and credit risk. Smaller firms should be more likely to rely on relationship lending (Huber, 2021), as they do not have the same access to credit as larger firms. Similarly, younger firms do not have a long credit history, and thus soft information may be more valuable for them. With respect to credit risk, two contrasting theories can be formulated. On the one hand, bank managers can engage in cream-skimming (Di Maggio and Yao, 2021), and so they may be more likely to bring with them less risky clients. On the other hand, if soft information is more valuable for riskier clients, then managers may be more

likely to bring with them riskier clients.

We start with firm size. We follow Eurostat cathegorization and define four bins of firm size: micro firms, with less than 10 employees, small firms, that have less than 50 employees (thus including the micro firms), medium firms, that have less than 250 employees (thus including small and micro firms), and large firms, that have more than 250 employees. We then interact the treatment with dummies that capture the size of the firms. Table 5 shows the results.

Table 5. Heterogeneity by firm size

| | Credit indicator | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Inflow | 0.019*** | 0.021*** | 0.024*** | 0.028*** |
| | (0.006) | (0.008) | (0.008) | (0.008) |
| Micro × Inflow | 0.012 | | | |
| | (0.008) | | | |
| Small × Inflow | | 0.007* | | |
| | | (0.004) | | |
| Medium × Inflow | | | -0.005 | |
| | | | (0.008) | |
| Big × Inflow | | | | -0.030*** |
| | | | | (0.010) |
| $R^2$ | 0.793 | 0.793 | 0.793 | 0.793 |
| Observations | 97,198,970 | 97,198,970 | 97,198,970 | 97,198,970 |
| Firm-Time fixed effects | ✓ | ✓ | ✓ | ✓ |
| Branch-Time fixed effects | ✓ | ✓ | ✓ | ✓ |
| Branch-Firm fixed effects | ✓ | ✓ | ✓ | ✓ |

*Notes:* DiD estimates for $\beta$ in equation 1 for different types of firms. The dependent variable is the credit indicator. Standard errors are clustered at the branch-firm level. Column (1) includes an interaction between $\beta$ and a dummy for the firm being micro (less than 10 employees), column (2) includes an interaction with a dummy for the firm being small (less than 50 employees), column (3) includes an interaction with a dummy for the firm being medium (less than 250 employees), and column (4) includes an interaction with a dummy for the firm being big (more than 250 employees). Moves are defined as relocations of managers from one bank group to another. To keep consistency with the sample definition, only moves that occur within the same

province are considered. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Although with a noisy estimate, the interaction term for micro firms is positive, suggesting that when firms are so little, they need a more direct relationship with the bank manager, and so they are more likely to follow her. As firms grow, the differential effect decreases: the interaction is negative for medium and large firms, and for large firms the total effect is close to zero ($0.028 - 0.030$).

Table 6. Heterogeneity by firm age and credit risk

| | Credit indicator | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Inflow | 0.023*** | 0.029*** | 0.025*** | 0.024*** |
| | (0.007) | (0.009) | (0.007) | (0.008) |
| Young × Inflow | 0.033*** | | | |
| | (0.012) | | | |
| Old × Inflow | | -0.009 | | |
| | | (0.006) | | |
| Safe × Inflow | | | -0.002 | |
| | | | (0.004) | |
| Risky × Inflow | | | | -0.0006 |
| | | | | (0.004) |
| $R^2$ | 0.793 | 0.793 | 0.793 | 0.793 |
| Observations | 97,198,970 | 97,198,970 | 97,198,970 | 97,198,970 |
| Firm-Time fixed effects | ✓ | ✓ | ✓ | ✓ |
| Branch-Time fixed effects | ✓ | ✓ | ✓ | ✓ |
| Branch-Firm fixed effects | ✓ | ✓ | ✓ | ✓ |

*Notes:* DiD estimates for $\beta$ in equation 1 for different types of firms. The dependent variable is the credit indicator. Standard errors are clustered at the branch-firm level. Column (1) includes an interaction with a dummy for the firm being young (less than 5 years old), column (2) includes an interaction with a dummy for the firm being old (more than 10 years old), column (3) includes an interaction with a dummy for the firm being safe (credit score below 4 in a 1-10 scale), and column (4) includes an interaction with a dummy for the firm being risky (credit score above 6 in a 1-10 scale). Moves are defined as relocations of managers from one bank group to another. To keep consistency with the sample definition, only moves that occur within the same province

are considered. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 6 shows the results for the age and credit risk of the firms. We define two bins of firm age: young firms, that had been founded in the 5 years before the manager moved, and old firms, that had been founded more than 10 years before the manager moved. As for risk, the Italian credit registry provides a credit risk score for each firm, which is computed every year by Cebi-Cerved (Rodano et al., 2018). The score ranges from 1 to 10, where 1 is the lowest risk and 10 is the highest risk. We take the Cebi-Cerved definitions and label as low risk firms that have a score lower or equal than 4, and as high risk firms that have a score higher or equal to 7.

The estimate for the interaction term with young firms is positive and significant, suggesting that younger firms are far more likely to follow their manager when she moves, as the increase is of the same order of magnitude of the baseline effect. On the other hand, the interaction for older firms is small and not significant, so that their total effect (about 0.020) is close to the baseline. With respect to credit risk, both estimates are non significant, and they are very close to zero. We then do not find evidence of cream-skimming, nor that riskier firms are more likely to follow their manager. Notice however that these results do not imply that - within the same risk category - the manager is not able to tell apart riskier and less risky firms, and so to bring with her the less risky ones. Results on the performance of the loans that are brought by the manager are provided in Section 7 and show that these loans perform better than the average loan of the welcoming bank. What is more, results in this subsection hint that bank manager moves can be effective in increasing the amount of information in the economy, as they help smaller and younger firms to access credit.

### 6.3 Manager characteristics

The third element in our analysis - for which we have unique data - is the bank manager. Relationship lending literature has speculated on the cultural (Fisman et al., 2017) and geographical (Amberg and Becker, 2024) proximity between the branch and the borrower, but has not been able to investigate individual characteristics of the bank manager, due to lack of individual level data. Frame et al. (2025) use loan level data with the identity of the loan officer to measure the impact of loan officer ethnicity on home mortgages in the US, but they do not follow loan officers when they move to a new bank, and so they cannot investigate the portability of the loan officer's clients. Theoretical literature (Gennaioli et al., 2015) suggests that trust

may be a key element in banking activity. Guided by this intuition, we expect older managers to be more effective in bringing their clients with them, as they may have built a larger relationship capital with their clients. At the same time, in smaller branches, managers may have a more direct relationship with their clients, and so being more effective in bringing them with them. Indeed, if a manager was working in a larger branch, with more managers, although possibly interacting with all the firms that were receiving credit from the branch, she may have had less chances to build a relationship with all of them, and maybe she focused on interacting with a subset of them. We then interact the treatment effect in equation (1) with the age of the manager, and with the number of managers in the branch. Table 7 shows the results for the age of the manager.

Table 7. Heterogeneity by managers' age

| | Credit indicator | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Manager younger than 45 | 0.022*** | | | |
| | (0.007) | | | |
| Manager older than 45 | | 0.026** | | |
| | | (0.012) | | |
| Manager younger than 55 | | | 0.022*** | |
| | | | (0.008) | |
| Manager older than 55 | | | | 0.043** |
| | | | | (0.020) |
| $R^2$ | 0.786 | 0.786 | 0.786 | 0.786 |
| Observations | 44,681,890 | 44,681,890 | 44,681,890 | 44,681,890 |
| Dependent variable mean | 0.013 | 0.013 | 0.013 | 0.013 |
| Firm-Time fixed effects | ✓ | ✓ | ✓ | ✓ |
| Branch-Time fixed effects | ✓ | ✓ | ✓ | ✓ |
| Branch-Firm fixed effects | ✓ | ✓ | ✓ | ✓ |

*Notes:* DiD estimates for $\beta$ in equation 1 with heterogeneity by managers' age. Column (1) includes an interaction between $\beta$ and a dummy for the manager being younger than 45 years old at the time of the move. In column (2), the interaction is with a dummy for the manager being older than 45 years old. Columns (3) and (4) repeat the exercise with a cutoff at 55

years old. Notice that in all specifications there is no baseline effect, so the estimates can be interpreted as the total effect for each group. Moves are defined as relocations of managers from one bank group to another. To keep consistency with the sample definition, only moves that occur within the same province are considered. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Columns (1) and (3) show results for young managers, being respectively younger than 45 and 55 years old. Comparing these estimates with the results in columns (2) and (4), that show the results for older managers, we find that older managers are more effective in bringing their clients with them. Specifically, the estimate for managers older than 55 years old is 0.043, which is almost double the effect of 0.022 for managers younger than 55 years old.

Table 8 shows the results for the number of managers in the branch. Notice that this regression can be also interpreted as a robustness check of our results, as we are able to test the sensitivity of our estimates to the choice of the threshold for the number of firms in a branch. Indeed, when defining our dataset, we confined ourselves at small branches (with less than 150 firms) in order to be able to have a reliable matching between the manager and the firms. There is a clear trade-off between increasing the precision of our estimates (confining ourselves to smaller branches) and considering a sizable number of moves. However, Appendix sections F.2 and F.1 show that our results are robust to changing the threshold to 50 and 100 firms.

Columns (1) and (3) show results for small branches, respectively with 3 or less, and 5 or less managers. As it can be seen comparing these estimates with the results in columns (2) and (4), managers coming from smaller branches are more effective in bringing their clients with them by an order of magnitude - i.e. 0.12 vs 0.013 comparing columns (1) and (2). Estimates in column (3), when small branches are defined by having 5 or less managers, are smaller in magnitude (0.085 as compared to 0.12 in column (1)). We can rationalize this fact both with the measurement argument (i.e. the more managers, the less likely it is that we matched a manager with only her clients), and with the span of control argument (i.e. in larger branches, managers interact less often with their clients, and so they are less effective in bringing them with them).

Table 8. Heterogeneity by number of managers

| | Credit indicator | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| From $\leq$ 3 managers branch | 0.120*** | | | |
| | (0.032) | | | |
| From $>$ 3 managers branch | | 0.013** | | |
| | | (0.006) | | |
| From $\leq$ 5 managers branch | | | 0.085*** | |
| | | | (0.024) | |
| From $>$ 5 managers branch | | | | 0.012* |
| | | | | (0.006) |
| $R^2$ | 0.786 | 0.786 | 0.786 | 0.786 |
| Observations | 44,681,890 | 44,681,890 | 44,681,890 | 44,681,890 |
| Dependent variable mean | 0.013 | 0.013 | 0.013 | 0.013 |
| Firm-Time fixed effects | ✓ | ✓ | ✓ | ✓ |
| Branch-Time fixed effects | ✓ | ✓ | ✓ | ✓ |
| Branch-Firm fixed effects | ✓ | ✓ | ✓ | ✓ |

*Notes:* DiD estimates for $\beta$ in equation 1 for different number of managers in the branch the loan officer moved from. Column (1) includes moves from branches with one manager, column (2) includes moves from branches with one or two managers, and column (3) includes moves from branches with one, two, or three managers. Only firms and branches that are present in the data in all years are included. Moves are defined as relocations of managers from one bank group to another. To keep consistency with the sample definition, only moves that occur within the same province are considered. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

However, consider that even in the presence of measurement error, it would bias our estimates towards zero. Therefore, the fact that our baseline results in Table 2 are positive and significant, and that results in this section - as well as in Appendix sections F.2 and F.1 - yield even larger estimates for smaller branches, is consistent with our baseline estimates being a lower bound of the true effect.

### 6.4 Loan size

Table 9. Heterogeneity by loan size

| | Credit indicator | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Former loan $<$ 50k | 0.024** | | | |
| | (0.011) | | | |
| Former loan $<$ 100k | | 0.025** | | |
| | | (0.010) | | |
| Former loan $<$ 500k | | | 0.032*** | |
| | | | (0.010) | |
| Former loan $\geq$ 500k | | | | 0.011* |
| | | | | (0.006) |
| $R^2$ | 0.793 | 0.793 | 0.793 | 0.793 |
| Observations | 97,198,970 | 97,198,970 | 97,198,970 | 97,198,970 |
| Firm-Time fixed effects | ✓ | ✓ | ✓ | ✓ |
| Branch-Time fixed effects | ✓ | ✓ | ✓ | ✓ |
| Branch-Firm fixed effects | ✓ | ✓ | ✓ | ✓ |

*Notes:* DiD estimates for $\beta$ in equation 1 for different loan sizes of the borrower's former loan with the relocating banker. Column (1) includes an interaction term with loans smaller than 50 thousand euros, column (2) includes an interaction term with loans smaller than 100 thousand euros, column (3) includes an interaction term with loans smaller than 500 thousand euros, and column (4) includes an interaction term with loans greater than 500 thousand euros. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

We then look at the role of the size of the relationship that the firm had before the manager moved. We define four bins of loan size: below 50 thousand euros, below 100 thousand euros (including the previous bin), below 500 thousand euros (including the previous bins), and above 500 thousand euros. A priori, it is not clear how relationship size should affect the portability of the manager's clients. Smaller loans may be more likely to be relationship loans, where the presence of hard information is limited, and so managers can transfer them more easily. At the same time, managers may have a stronger incentive to bring with them larger loans, as they may be more remunerated for them. Table 9 shows that the effect rises from 0.025 for former loans below 100 thousand euros to 0.032 for loans below 500 thousand euros, around one and a half times the baseline effect of 0.023 in Table 2. Since the bins are cumulative, this implies

that mid-size loans (between 100 and 500 thousand euros) are the most likely to follow their manager. Smaller loans are as portable as the baseline effect, while larger loans are less portable, with a point estimate of 0.011 that is statistically significant only at the 10% level.

### 6.5 Competition

The last dimension of heterogeneity we investigate on relates to the level of competition in the local banking market. Indeed, bank managers are active in a local labor market, and their skills encompass the knowledge of the local credit market, and in particular the firms they had relationships with. Therefore worker relocations to local competitors may alter the competitive landscape of that market, especially if managers are able to bring their clients with them. We then ask if the portability of the manager's clients is affected by the level of competition in the local banking market. We measure competition by means of the Herfindahl-Hirschman Index (HHI), computed at the province level, using as weights the total amount of loans that each bank group has in that province. We exclude provinces where concentration index is above 0.25, as they are considered to be highly concentrated markets, and we also exclude provinces where the index is below 0.1, as competition is too fragmented. Appendix figure A5 shows the HHI distribution in our sample.

On this subset of provinces, we define four quartiles of HHI, and we interact the treatment effect in equation (1) with dummies that capture the quartile of the province where the manager moved to. A priori, it is not clear how competition should affect the portability of the manager's clients. On the one hand, in more competitive markets, hiring managers with a strong local network may be more appealing for banks, that use the knowledge of the manager to poach clients from competitors. On the other hand, in more concentrated markets, firms have fewer alternatives, and so they may be more likely to follow their manager when she moves.

Table 10. Heterogeneity by local market concentration

| | Credit indicator | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| Bottom HHI quartile × Inflow | 0.100*** | | |
| | (0.034) | | |
| Below median HHI × Inflow | | 0.018*** | |
| | | (0.006) | |
| Below 75pct HHI × Inflow | | | 0.023** |
| | | | (0.009) |
| $R^2$ | 0.788 | 0.788 | 0.788 |
| Observations | 27,124,990 | 27,124,990 | 27,124,990 |
| Dependent variable mean | 0.013 | 0.013 | 0.013 |
| Firm-Time fixed effects | ✓ | ✓ | ✓ |
| Branch-Time fixed effects | ✓ | ✓ | ✓ |
| Branch-Firm fixed effects | ✓ | ✓ | ✓ |

*Notes:* DiD estimates for $\beta$ in equation 1 for different types of bank moves. Column (1) includes an interaction term with moves in provinces that are in the bottom quartile of the HHI distribution - i.e. the ones with in more competitive environments. Column (2) includes an interaction term with moves in provinces that are below the median of the HHI distribution, and column (3) includes an interaction term with moves in provinces that are below the 75th percentile of the HHI distribution - i.e. increasing the level of concentration in the market. The HHI is computed at the province level using the total amount of loans granted by each bank group in the province in 2018. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 10 shows the results. Notice that this regression features only the interaction term, therefore the estimates in columns (1) to (3) have to be interpreted as the total effect of the manager's move. Furthermore, as for firm or loan size, bins (in this case quartiles) have been defined in a progressive way, such that each bin includes the previous one. Portability is highest in the most competitive markets: in the bottom quartile of the HHI distribution, the estimate is 0.100, more than four times the baseline effect of 0.023 in Table 2. Widening the bin lowers the estimate to 0.018 below the median and 0.023 below the $75^{th}$ percentile, so that outside the most competitive markets the effect is close to, or below, the baseline. This pattern is consistent with the first channel discussed above: in more competitive markets, banks hire managers with a strong local network to poach clients from their competitors.

# 7 Loan terms and performance

We analyse the terms and performance of loans granted to switcher firms. Differently from the previous sections, here we do not use the expanded panel of potential firm-branch pairs: since loan terms and performance are only defined for realized loans, the analyses in this section are run on the credit relationships observed in the credit registry. We want to understand the incentives for switching firms to move their loan activity to a new bank. The key driver of the decision to switch banks is likely to be the interest rate that firms pay on their loans. We observe that switchers pay lower interest rates on their loans after the switch, compared to what they were paying before the switch. This result holds controlling for firm characteristics, such as riskiness. Switchers do not receive preferential treatment from the new branch: the interest rate they pay is the same as that of other new borrowers in that branch. This is consistent with Beraldi (2025), who finds that banks do not price risk when they renew loans to existing clients, but they do when they grant loans to new clients. In this case we are able to show that when the manager changes bank, her clients obtain better terms than those they had at their previous bank, even when controlling for riskiness.

At the same time, banks are happy to lend to switchers, as long as their repayment performance is better than that of other firms. We find that switchers have lower NPL rates than other firms, even when controlling for riskiness. This suggests that bankers have superior information about their clients, and they use it to award loans to the ones that - within each riskiness category - have a higher repayment probability.

## 7.1 Interest rates

Interest rates are only available at the bank group level, so we estimate the following regression at that level:

$$i_{g(b)ft} = \beta \text{Switcher}_{g(b)ft} + \gamma \log(1 + \text{credit}_{g(b)ft}) + \mathbf{X}_{g(b)ft} + \epsilon_{g(b)ft} \tag{5}$$

where $i_{g(b)ft}$ is the average interest rate on loans granted to firm $f$ by bank group $g(b)$ over a two-year period $t$. $\text{Switcher}_{g(b)ft}$ is an indicator equal to one if the loan was granted after the firm followed its loan officer to bank group $g(b)$ in year $t$ (or later). $\text{credit}_{g(b)ft}$ is the average amount of credit granted to the firm by the bank group over the two-year period $t$. $\mathbf{X}_{g(b)ft}$ is a vector of controls, including firm characteristics (age, size, riskiness), bank group, year and banker fixed effects. Standard errors are

clustered at the firm level.

We estimate equation (5) using five different comparison groups, each corresponding to a different question. First, we ask if switchers pay less when they open a new relationship with their banker, compared to what they were paying in all their other loans (before and after the switch). Second, we ask if switchers would have been better off simply shopping around, rather than following their banker to a new bank. We do this by comparing switchers to their fellow portfolio firms (i.e., firms that could have switched, but did not) and measuring if - when those firms get new loans - switchers are able to get better terms. The third comparison is similar to the second one, but we compare switchers to the old relationships of portfolio firms (i.e., relationships that existed before the banker moved to a new bank). It serves to understand if switchers would have been better off staying with one of their old relationships, rather than following their banker to a new bank. After having considered the possible choices that switchers had, we then ask if they receive some form of preferential treatment within the new branch of the loan officer. We do this by comparing switchers to the new relationships of the new branch where the loan officer moved. Finally, we zoom in on the switchers trade-off, comparing the interest rates they pay after the switch to the interest rates they were paying with their old bank (from which the loan officer moved away).

Table 11 reports the results of the five comparisons.

Table 11. Interest rates performance

| | Interest rate | | | | |
|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) |
| Switcher | -0.505* | -0.657 | -0.289* | 0.050 | -0.926*** |
| | (0.264) | (0.699) | (0.172) | (0.144) | (0.195) |
| Log. average credit (2y) | -0.555*** | -0.669*** | -0.716*** | -0.268*** | -0.773*** |
| | (0.019) | (0.004) | (0.003) | (0.004) | (0.097) |
| $R^2$ | 0.198 | 0.197 | 0.188 | 0.123 | 0.311 |
| Observations | 6,643 | 23,609 | 35,585 | 68,555 | 1,387 |
| Dependent variable mean | 2.55 | 2.53 | 2.92 | 2.60 | 2.93 |

*Notes:* Estimates for the effect on interest rates of switching with the loan officer. The dependent variable is the average interest rate on loans over a two-year period. The key independent variable is an indicator equal to one if the firm switched with the loan officer. All regressions include as

controls the log of average credit granted over a two-year period, firm characteristics (age, size, riskiness), bank group, year and banker fixed effects. Standard errors are clustered at the firm level. In all columns, the treated group is defined by the loans that switchers receive having followed their loan officer to a new bank. What varies across columns is the definition of the control group. In column (1) the control group is made of all other loans of switchers (before and after the switch). In column (2) the control group is made of all the new relationships of portfolio firms, that did not switch with the banker. In column (3) the control group is made of all the old relationships of portfolio firms, that did not switch with the banker (i.e., relationships that existed before the banker moved to a new bank). In column (4) the control group is made of all the new relationships of the new branch of the loan officer, that were not switchers. In column (5) the control group is made of the old relationships of switchers, i.e., the loans that switchers had with their old branch before they switched with the banker. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Column (1) shows that switchers pay on average 0.5 percentage points less than what they were paying in all their other loans (before and after the switch). The result is statistically significant at the 10% level, and economically sizable, as the average interest rate in the sample is 2.55%. It suggests that - within firms that switch with their banker - following the banker to a new bank allows them to get better terms. Column (2) shows that switchers have been slightly better off following their banker to a new bank, rather than simply shopping around. However, the effect is not statistically significant. Column (3) shows that switchers pay on average 0.3 percentage points less than what portfolio firms were paying in their old relationships (i.e., relationships that existed before the banker moved to a new bank). The effect is statistically significant at the 10% level, and goes in the same direction as column (1). In column (4) we find no difference in the interest rates paid by switchers and by the new relationships of the new branch. We interpret this result as evidence that bankers do not give preferential treatment to switchers, as opposed to new relationships of the new branch. Finally, column (5) shows that switchers pay on average 0.9 percentage points less than what they were paying with their old bank (with the same loan officer), suggesting that the switch opens up new opportunities for them.

We interpret these results as evidence that banker movements create value for their clients, opening up new opportunities that would not have been available otherwise. At the same time, there is no evidence of rent-sharing between bankers and their clients, as switchers do not pay lower interest rates than new relationships of the new branch. In Appendix C we breakdown the effects by type of loan (self-liquidating loans, credit lines and term loans), showing that most of the variation is driven by credit lines.

### 7.2 Non-performing loans

Once we established that firms have an incentive to follow their banker to a new bank, we want to understand if banks can somehow benefit from the knowledge that bankers have about their clients. We measure default with an indicator for non-performing loans (NPL). We look at ex post performance of loans granted to switchers, estimating if they have lower NPL rates than other firms. If managers have superior information about their clients, they should be able to identify and award loans to the more deserving ones, although hard information (e.g., credit scores) reports them as equally risky. We estimate the following regression:

$$\text{NPL}_{bft} = \beta \text{Switcher}_{bft} + \sum_{j=1}^{3} \gamma_j \log(1 + \text{credit}_{fbjt}) + \mathbf{X}_{bft} + \epsilon_{bft} \tag{6}$$

where $\text{NPL}_{bft}$ is an indicator equal to one if firm $f$ has at least one non-performing loan (NPL) in year $t$ with branch $b$ (the NPL data are available at the branch level). $\text{Switcher}_{bft}$ is - as before - an indicator equal to one if the loan was granted after the firm followed its loan officer to branch $b$ in year $t$ (or later). $\text{Log}(1+\text{credit}_{fbjt})$ is the log of one plus the amount of credit granted to firm $f$ by branch $b$ in year $t$, broken down by type of loan $j$ (self-liquidating loans, credit lines and term loans). $\mathbf{X}_{bft}$ is a vector of controls, including firm characteristics (age, size, riskiness), bank group-year, banker and municipality fixed effects. Notice that we are able to include even a finer set of fixed effects with respect to Equation (5). These fixed effects, that control for branch activity at the municipality level, are now possible because the dataset on non-performing loans is available at the branch level. Standard errors are clustered at the firm level.

Similarly to what we did for interest rates, we estimate Equation (6) using four different comparison groups, each corresponding to a different question. First, we ask if relationships originated by switchers with their own banker have lower NPL rates than all their other loans (before and after the switch). We then compare switchers to their fellow portfolio firms that have shopped around (i.e., firms that could have switched, but did not) and measure if - when those firms get new loans - switchers have lower NPL rates. The third option for switchers would have been to stay with one of their old relationships, rather than following their banker to a new bank. We do this by comparing switchers to the old relationships of portfolio firms (i.e., relationships that existed before the banker moved to a new bank). Finally, we ask if switchers

have lower NPL rates than the new relationships of the new branch where the loan officer moved.

Table 12. NPL probability performance

| | NPL probability | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Switcher | -0.014** | -0.011 | -0.020* | -0.043** |
| | (0.006) | (0.007) | (0.011) | (0.019) |
| $R^2$ | 0.115 | 0.101 | 0.055 | 0.048 |
| Observations | 13,320 | 45,700 | 65,195 | 187,389 |
| Dependent variable mean | 0.017 | 0.027 | 0.015 | 0.016 |

*Notes:* Estimates for the effect on NPL probability of switching with the loan officer. The dependent variable is the probability of non-performing loans (NPL) over a two-year period. The key independent variable is an indicator equal to one if the firm switched with the loan officer. All regressions include as controls the log of credit granted (broken down by type of loan), firm characteristics (age, size, riskiness), municipality, bank group-year and banker fixed effects. Standard errors are clustered at the firm level. In all columns, the treated group is defined by the loans that switchers receive having followed their loan officer to a new bank. What varies across columns is the definition of the control group. In column (1) the control group is made of all other loans of switchers (before and after the switch). In column (2) the control group is made of all the new relationships of portfolio firms, that did not switch with the banker. In column (3) the control group is made of all the old relationships of portfolio firms, that did not switch with the banker (i.e., relationships that existed before the banker moved to a new bank). In column (4) the control group is made of all the new relationships of the new branch of the loan officer, that were not switchers. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 12 reports the results of the four comparisons. Column (1) shows that switchers have on average a 1.4 percentage points lower probability of having at least one non-performing loan in a year, compared to all their other loans (before and after the switch). The result - which is statistically significant at the 5% level - shows us that when firms follow their banker to a new bank, they tend to perform better with respect to the other credit relationships they have. From column (2) we see that switchers also have a lower NPL rate than their fellow portfolio firms that have shopped around, although the effect is smaller and not statistically significant. Column (3) compares newly originated relationships of switchers to the old relationships of portfolio firms, showing that switchers have a 2 percent lower NPL rate than those

old relationships. The effect is statistically significant at the 10% level. Column (4) is the most interesting one from the point of view of banks. It shows that switchers have a 4.3 percentage points lower NPL rate than the new relationships of the new branch. This result holds controlling for riskiness and other firm characteristics, and it is statistically significant at the 5% level. We interpret it as evidence that bankers can benefit their new bank, as they have superior information about their clients. In principle, there should be no difference in the quality of monitoring between switchers and new relationships of the new branch. However, switchers are significantly less likely to have an NPL (95% confidence interval: from −8.0 to −0.6 percentage points). Table A12 in Appendix D shows that the effect is persistent, as it holds even two years after the loan has been granted.

In Appendix D we also reproduce the results of Table 12 interacting the switcher indicator with an indicator for riskiness. The scope of the exercise is to understand if bankers are able - within the bucket of riskier firms - to identify and award loans to the ones that have a higher repayment probability. As it can be seen in Table A13, there is no significant effect in columns (1), (3) and (4). However, in column (2) we find that risky switchers have a 3.7 percentage points lower NPL rate than their fellow risky portfolio firms that have shopped around. Put it differently, from the perspective of a risky firm, choosing to follow its banker to her new bank yields a significant improvement in the probability of having an NPL, compared to simply shopping around. This result can be driven both by selection (bankers are able to tell which risky firms are less likely to become non-performing) and pricing (as it was shown in the previous section, switchers pay lower interest rates). However, we regard that in both cases the key driver is the superior information that bankers have about their clients, which allows them to identify and award loans to the more deserving ones, even when hard information (e.g., credit scores) reports them as equally risky.

### 7.3 Credit amounts and reallocation

Beyond the terms of the contract, we ask what happens to the *quantity* of credit when a firm follows its banker. We first look within movers, comparing each follower's credit before and after the move (Table 13). Column (1) shows that the relationship the firm opens at the new branch is about 7% smaller than the one it had at the old branch, while column (2) shows that the firm's *total* credit grows by about 87% in the move year (a coefficient of 0.626 on the log of total credit) — credit is redirected and expanded, not merely relocated. Column (3) measures how much credit remains

at the old branch after the move.[11]

We then take a cross-sectional view (Table 14), comparing relationships that followed a manager against four different control sets. Relative to the follower firm's own book the followed relationship is smaller (column 1), and relative to other new clients at the destination branch it is also smaller (column 4); but relative to portfolio firms' other new relationships it carries more total credit (column 2), and it is indistinguishable from portfolio firms' old stock (column 3). The by-loan-type breakdowns (Appendix E) are consistent with a participation margin — followed relationships are present in more loan categories rather than larger within any single one.

Table 13. Credit reallocation around a manager's move

| | New vs old branch relationship | Firm total credit | Old-branch retention |
|---|---|---|---|
| | (1) | (2) | (3) |
| After move | -0.070* | 0.626*** | |
| | (0.037) | (0.087) | |
| $R^2$ | 0.926 | 0.853 | |
| Observations | 1,486 | 1,482 | |
| Dependent variable mean | 12.8 | 13.8 | |

*Notes:* Before/after comparisons around the year a manager relocates, on the sample of realized moves (firm–transitions where the firm opened an original relationship at the manager's new branch). Each column is a separate regression of the outcome on an *After move* indicator, saturating with pair (firm × transition) fixed effects and firm characteristics (age, size, riskiness); standard errors clustered at the firm level. In column (1) the dependent variable is the log of credit granted, comparing the firm's relationship at the old branch (before) with the relationship the manager opens at the new branch (after). In column (2) it is the log of the firm's total credit across all banks and branches, at the year before vs the year of the move. In column (3) it is an indicator that any credit remains at the old branch after the move (extensive-margin retention); *[coefficient pending re-estimation on the current data].* *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 14. Loan amounts of followed relationships

[11] Under the current specification this is the extensive-margin probability that any credit remains at the old branch; the estimate is pending re-estimation on the current data. The by-loan-type breakdowns for all three comparisons are reported in Appendix E.

| | Log credit granted (total) | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Manager inflow | -0.401* | 0.556* | 0.002 | -0.353* |
| | (0.230) | (0.315) | (0.096) | (0.196) |
| $R^2$ | 0.562 | 0.465 | 0.395 | 0.277 |
| Observations | 12,690 | 44,795 | 64,275 | 187,256 |
| Dependent variable mean | 13.5 | 13.1 | 13.0 | 12.3 |

*Notes:* Cross-sectional comparisons of the loan amounts on relationships that followed a manager (*Manager inflow* = 1 if a followed manager relocated to that firm–branch on or before the credit year) against different control sets. The dependent variable is the log of credit granted (total). All regressions use the saturated specification: municipality, bank group-year and manager fixed effects plus firm characteristics (age, size, riskiness); standard errors clustered at the firm level. In column (1) the control group is the follower firm's own other relationships (any year). In column (2) it is portfolio firms' other new relationships (formed at or after the move). In column (3) it is portfolio firms' old, pre-move relationships. In column (4) it is other new clients at the branches that received a followed manager. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

# 8 Conclusion

In this paper, we show that when bank managers move, they bring their clients with them. By building personal relationships and acquiring soft information, managers use this knowledge to benefit the bank that hires them.

A manager's move increases a firm's likelihood of establishing a credit relationship with the new bank by 3.7 percentage points — almost quadrupling the baseline matching probability. Using loan application data, we identify two key channels: first, managers increase the likelihood that firms they know apply for loans at their new bank; second, they improve firms' chances of loan approval. These effects are stronger when the manager is more experienced, when she is moving from a larger bank to a smaller one, or when firms in her portfolio are smaller and younger. Furthermore, we find that managers improve credit conditions for their clients, as evidenced by lower interest rates with respect to their previous bank. Interestingly, firms that follow their manager do not pay lower interest rates with respect to other firms at the new bank, suggesting that there is no rent-sharing between the manager and her clients. Banks also appear to benefit from opening new relationships to the manager's clients,

as these firms are 4 percentage points less likely to report an NPL than other firms at the bank.

We believe our findings show how still relevant soft information is in credit allocation, even in an era of increasing reliance on algorithms and big data. They also highlight a new channel of bank competition, through the hiring of bank managers. This opens up a new avenue for policy intervention in the banking sector, as regulators may consider bank managers as a separate actor in the credit market, with their own incentives and objectives. However, we believe our findings are only a first step in understanding the broader implications of bank manager mobility for banks, managers, and firms.

Specifically, we identify three main areas for future research. First, what are the implications for the bank? Besides the lower NPL rates we document, does the bank benefit from the manager's clients in terms of profitability, growth, or market share? Second, what are the implications for the manager? Does she benefit from bringing her clients along in terms of career advancement or compensation? Third, what are the implications for the firms? Do firms that follow their manager experience better survival, growth, or profitability outcomes? We believe that - especially for policy purposes - these are important questions that deserve further attention both from academic research and from regulators.

## References


**Acabbi, Edoardo Maria, Ettore Panetti, and Alessandro Sforza**, "The Financial Channels of Labor Rigidities: Evidence from Portugal," 2024.

**Albertazzi, Ugo, Margherita Bottero, and Gabriele Sene**, "Information externalities in the credit market and the spell of credit rationing," *Journal of Financial Intermediation*, April 2017, *30*, 61–70.

**Alessandri, Piergiorgio and Margherita Bottero**, "Bank Lending in Uncertain Times," *European Economic Review*, 2020, *128*, 103503.

**Amberg, Niklas and Bo Becker**, "Banking Without Branches," 2024.

**Babina, Tania, Greg Buchak, and Will Gornall**, "Customer Data Access and Fintech Entry: Early Evidence from Open Banking," *SSRN Electronic Journal*, 2022.

**Bandiera, Oriana, Andrea Prat, Stephen Hansen, and Raffaella Sadun**, "CEO Behavior and Firm Performance," *Journal of Political Economy*, 2020.

**_ , Iwan Barankay, and Imran Rasul**, "Social Connections and Incentives in the Workplace: Evidence From Personnel Data," *Econometrica*, 2009, *77* (4), 1047–1094.

**Barone, Guglielmo, Fabiano Schivardi, and Enrico Sette**, "Interlocking Directorates and Competition in Banking," *Journal of Finance*, 2024, *Forthcoming.*

**Bell, Brian and John Van Reenen**, "Bankers and Their Bonuses," *The Economic Journal*, February 2014, *124* (574), F1–F21.

**Beraldi, Francesco**, "Banking Relationships and Loan Pricing Disconnect," 2025.

**Bloom, Nicholas, Benn Eifert, Aprajit Mahajan, David McKenzie, and John Roberts**, "Does Management Matter? Evidence from India*," *The Quarterly Journal of Economics*, February 2013, *128* (1), 1–51.

**Bolton, Patrick, Javier Freixas, Leonardo Gambacorta, and Paolo Emilio Mistrulli**, "Relationship and Transaction Lending in a Crisis," *The Review of Financial Studies*, 2016, *29* (10).

**Bonfim, Diana, Gil Nogueira, and Steven Ongena**, ""Sorry, We're Closed" Bank Branch Closures, Loan Pricing, and Information Asymmetries," *Review of Finance*, July 2021, *25* (4), 1211–1259.

**Borusyak, Kirill, Xavier Jaravel, and Jann Spiess**, "Revisiting Event-Study Designs: Robust and Efficient Estimation," *Review of Economic Studies*, November 2024, *91* (6), 3253–3285.

**Boustanifar, Hamid, Everett Grant, and Ariell Reshef**, "Wages and Human Capital in Finance: International Evidence, 1970–2011*," *Review of Finance*, March 2018, *22* (2), 699–745.

**Branzoli, Nicola and Fulvia Fringuellotti**, "The Effect of Bank Monitoring on Loan Repayment," *SSRN Electronic Journal*, 2020.

**Bénabou, Roland and Jean Tirole**, "Bonus Culture: Competitive Pay, Screening, and Multitasking," *Journal of Political Economy*, April 2016, *124* (2), 305–370.

**Böhm, Michael J, Daniel Metzger, and Per Strömberg**, ""Since You're So Rich, You Must Be Really Smart": Talent, Rent Sharing, and the Finance Wage Premium," *Review of Economic Studies*, September 2023, *90* (5), 2215–2260.

**Chaisemartin, Clément De and Xavier D'Haultfœuille**, "Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects," *American Economic Review*, September 2020, *110* (9), 2964–2996.

**Cornell, Bradford and Ivo Welch**, "Culture, Information, and Screening Discrimination," *Journal of Political Economy*, June 1996, *104* (3), 542–571.

**Crawford, Gregory S., Nicola Pavanini, and Fabiano Schivardi**, "Asymmetric Information and Imperfect Competition in Lending Markets," *American Economic Review*, July 2018, *108* (7), 1659–1701.

**Degryse, Hans and Steven Ongena**, "Distance, Lending Relationships, and Competition," *The Journal of Finance*, February 2005, *60* (1), 231–266.

**Duquerroy, Anne, Clément Mazet-Sonilhac, Jean-Stéphane Mésonnier, and Daniel Paravisini**, "Bank Local Specialization," *SSRN Electronic Journal*, 2022.

**D'Andrea, Angelo, Marco Pelosi, and Enrico Sette**, "When Broadband Comes to Banks: Credit Supply, Market Structure, and Information Acquisition," *Journal of the European Economic Association*, 06 2026, *24* (3), 926–975.

**Eliason, Marcus, Lena Hensvik, Francis Kramarz, and Oskar Nordström Skans**, "Social connections and the sorting of workers to firms," *Journal of Econometrics*, 2023, *233* (2), 468–506.

**Fisman, Raymond, Daniel Paravisini, and Vikrant Vig**, "Cultural Proximity and Loan Outcomes," *American Economic Review*, February 2017, *107* (2), 457–492.

**Focarelli, Dario and Fabio Panetta**, "Are Mergers Beneficial to Consumers? Evidence from the Market for Bank Deposits," *American Economic Review*, August 2003, *93* (4), 1152–1172.

**Frame, W. Scott, Ruidi Huang, Erica Xuewei Jiang, Yeonjoon Lee, Will Shuo Liu, Erik J. Mayer, and Adi Sunderam**, "The Impact of Minority Representation at Mortgage Lenders," *The Journal of Finance*, April 2025, *80* (2), 1209–1260.

**Gao, Janet, Wenyu Wang, and Xiaoyun Yu**, "Big Fish in Small Ponds: Human Capital Mobility and the Rise of Boutique Banks," *Management Science*, 2024, *70* (11), 7829–7850.

**_ , _ , and Yufeng Wu**, "Human Capital Portability and Careers in Finance," *The Review of Financial Studies*, September 2024, *37* (9), 2732–2778.

**Gennaioli, Nicola, Andrei Shleifer, and Robert Vishny**, "Money Doctors," *The Journal of Finance*, February 2015, *70* (1), 91–114.

**Gobbi, Giorgio and Enrico Sette**, "Do Firms Benefit from Concentrating their Borrowing? Evidence from the Great Recession*," *Review of Finance*, April 2014, *18* (2), 527–560.

**Goedde-Menke, Michael and Peter-Hendrik Ingermann**, "Loan Officer Specialization and Credit Defaults," *Journal of Banking & Finance*, December 2023, p. 107077.

**Haselmann, Rainer, David Schoenherr, and Vikrant Vig**, "Rent Seeking in Elite Networks," *Journal of Political Economy*, 2018, *126* (4), 1638–1690. _eprint: https://doi.org/10.1086/697742.

**Hertzberg, Andrew, Jose Maria Liberti, and Daniel Paravisini**, "Information and Incentives Inside the Firm: Evidence from Loan Officer Rotation," *The Journal of Finance*, June 2010, *65* (3), 795–828.

**Huber, Kilian**, "Are Bigger Banks Better? Firm-Level Evidence from Germany," *journal of political economy*, 2021.

**Jasova, Martina, Caterina Mendicino, Ettore Panetti, Jose-Luis Peydro, and Dominik Supera**, "Monetary Policy, Labor Income Redistribution and the Credit Channel: Evidence from Matched Employer-Employee and Credit Registers," *SSRN Electronic Journal*, 2021.

**Keil, Jan and Steven Ongena**, "The demise of branch banking – Technology, consolidation, bank fragility," *Journal of Banking & Finance*, January 2024, *158*, 107038.

**Khwaja, Asim Ijaz and Atif Mian**, "Tracing the Impact of Bank Liquidity Shocks: Evidence from an Emerging Market," *American Economic Review*, September 2008, *98* (4), 1413–1442.

**Kosekova, Kamelia, Angela Maddaloni, Melina Papoutsi, and Fabiano Schivardi**, "Firm-bank relationships: a cross-country comparison," *CEPR Discussion Paper*, 2024, *19464*.

**Kramarz, F. and O. N. Skans**, "When Strong Ties are Strong: Networks and Youth Labour Market Entry," *The Review of Economic Studies*, July 2014, *81* (3), 1164–1200.

**Lazear, Edward P, Kathryn L Shaw, and Christopher T Stanton**, "The Value of Bosses," *Journal of Labor Economics*, 2015, *33* (4).

**Maggio, Marco Di and Vincent Yao**, "Fintech Borrowers: Lax Screening or Cream-Skimming?," *The Review of Financial Studies*, September 2021, *34* (10), 4565–4618.

**Malgieri, Cedomir and Luca Citino**, "Wage Contracts and Financial Frictions," 2025.

**Marco, Filippo De, Julien Sauvagnat, and Enrico Sette**, "Lending to Overconfident Borrowers," 2023.

**Minni, Virginia**, "Making the Invisible Hand Visible: Managers and the Allocation of Workers to Jobs," *The Quarterly Journal of Economics*, July 2026, *141* (3), 1871–1920.

**Nguyen, Hoai-Luu Q.**, "Are Credit Markets Still Local? Evidence from Bank Branch Closings," *American Economic Journal: Applied Economics*, January 2019, *11* (1), 1–32.

**Panetta, Fabio, Fabiano Schivardi, and Matthew Shum**, "Do Mergers Improve Information? Evidence from the Loan Market," *Journal of Money, Credit and Banking*, June 2009, *Vol. 41* (No. 4), pp. 673–709.

**Papoutsi, Melina**, "Lending Relationships in Loan Renegotiation: Evidence from Corporate Loans," 2025.

**Paravisini, Daniel, Veronica Rappoport, and Philipp Schnabl**, "Specialization in Bank Lending: Evidence from Exporting Firms," *The Journal of Finance*, August 2023, *78* (4), 2049–2085.

**Patault, Bérengère and Clémence Lenoir**, "Customer Capital Spillovers: Evidence from Sales Managers in International Markets," *American Economic Journal: Applied Economics*, October 2024, *16* (4), 404–443.

**Petersen, Mitchell A. and Raghuram G. Rajan**, "Does Distance Still Matter? The Information Revolution in Small Business Lending," *The Journal of Finance*, December 2002, *57* (6), 2533–2570.

**Philippon, Thomas and Ariell Reshef**, "Wages and Human Capital in the U.S. Finance Industry: 1909–2006*," *The Quarterly Journal of Economics*, November 2012, *127* (4), 1551–1609.

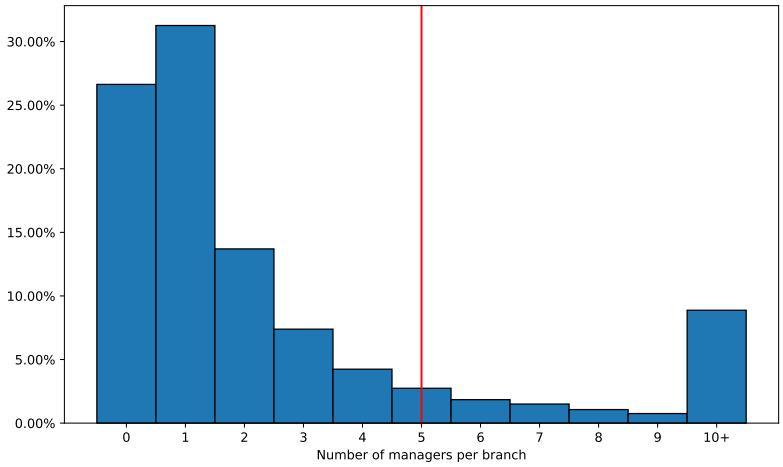
**Rodano, Giacomo, Nicolas Serrano-Velarde, and Emanuele Tarantino**, "Lending Standards over the Credit Cycle," *The Review of Financial Studies*, August 2018, *31* (8), 2943–2982.

**Stein, Jeremy C.**, "Information Production and Capital Allocation: Decentralized versus Hierarchical Firms," *The Journal of Finance*, October 2002, *57* (5), 1891–1921.

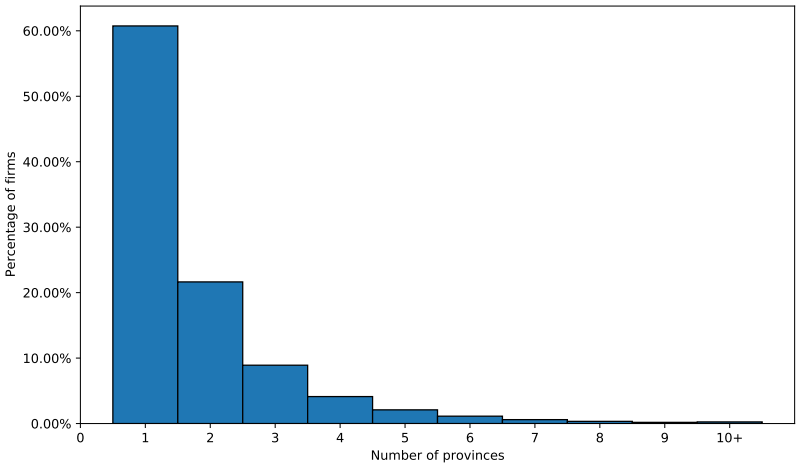
**Stiglitz, Joseph E and Andrew Weiss**, "Credit Rationing in Markets with Imperfect Information," *The American Economic Review*, 1981, *71* (3), 393–410.

**Sun, Liyang and Sarah Abraham**, "Estimating dynamic treatment effects in event studies with heterogeneous treatment effects," *Journal of Econometrics*, December 2021, *225* (2), 175–199.

# A Descriptive figures and tables




Appendix Figure A1. Distribution of the number of managers per branch.




Appendix Figure A2. Distribution of the number of provinces where firms have credit.

Appendix Table A1. Annualised revenue growth by firm size and leverage

| | Small ($\leq$ 10 empl.) | Big ($>$ 10 empl.) |
|---|---|---|
| *Panel A: 2009–2018* | | |
| High leverage | 2.02 | 1.97 |
| Low leverage | 1.72 | 1.49 |
| *Panel B: 2009–2013* | | |
| High leverage | 2.30 | 1.53 |
| Low leverage | 1.69 | 0.90 |
| *Panel C: 2013–2018* | | |
| High leverage | 3.30 | 2.85 |
| Low leverage | 3.04 | 2.56 |

*Notes:* Each cell reports the annualised mean revenue growth rate (in %) of Cerved firms, split by firm size (columns) and leverage (rows).

**Sample.** The sample includes all firms with non-missing and positive revenues in the start year, and with non-missing leverage and employment in the start year. The number of firms in each panel is 149,773 (2009–2018), 170,037 (2009–2013) and 200,785 (2013–2018).

**Growth rate.** For a firm with net revenues $R_{t_0}$ in the start year and $R_{t_1}$ in the end year, the annualised growth rate is $g = (R_{t_1}/R_{t_0})^{1/(t_1-t_0)} - 1$, where $t_1 - t_0$ is the number of years in the window (9, 4 and 5 years for the three panels respectively). The cell entry is the simple mean of $g$ across firms in the cell.

**Leverage.** Leverage is total debt towards banks (computed from the Credit Register) divided by total assets, measured in the start year. Firms are classified as *high* (*low*) leverage if their leverage is above (at or below) the sample median, computed within each window over the firms entering that panel: the median equals 0.430 (2009–2018), 0.423 (2009–2013), 0.370 (2013–2018).

**Size.** A firm is *small* if it has at most 10 total employees in the start year, and *big* otherwise.

# B Additional results

Appendix Table A2. DiD estimates for $\beta$, identified via branch closures

| | Credit indicator | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Inflow from branch closure | 0.184* | 0.217* | 0.183* | 0.216* |
| | (0.105) | (0.118) | (0.104) | (0.117) |
| $R^2$ | 0.772 | 0.785 | 0.773 | 0.786 |
| Observations | 44,681,890 | 44,681,890 | 44,681,890 | 44,681,890 |
| Dependent variable mean | 0.013 | 0.013 | 0.013 | 0.013 |
| Branch-Firm fixed effects | ✓ | ✓ | ✓ | ✓ |
| Branch-Year fixed effects | | ✓ | | ✓ |
| Firm-Year fixed effects | | | ✓ | ✓ |

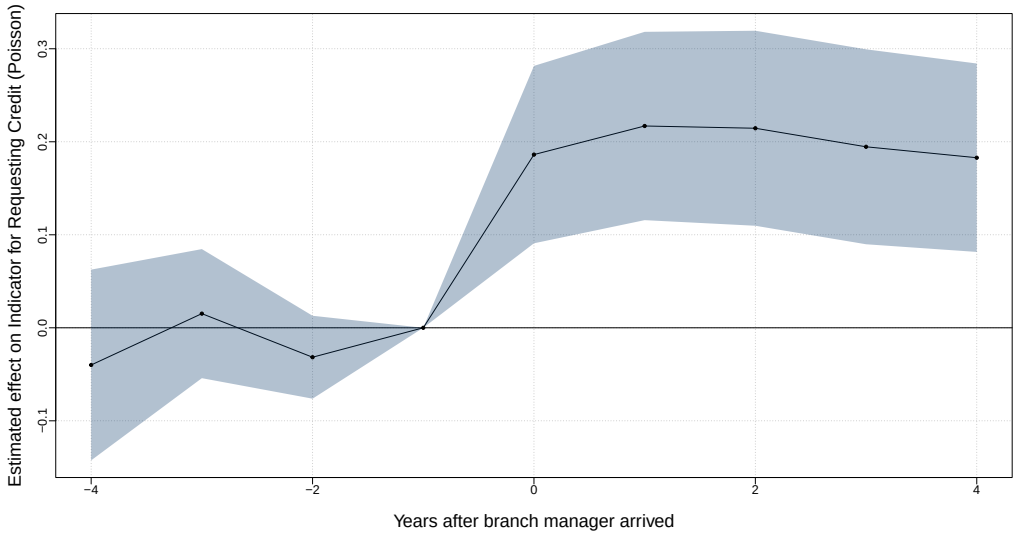
*Notes:* Estimates for the probability of a firm receiving credit from the branch where its previous loan officer has moved, after her branch has closed. Column (1) includes only branch fixed effects, column (2) includes branch and time fixed effects, column (3) includes branch and firm fixed effects, and column (4) includes all fixed effects. Standard errors are clustered at the bank group - firm level. The sample includes all potential firm - branch pairs within a province, for the period 2009-2018. Only firms and branches that are present in the data in all years are included. Moves are defined as relocations of managers from one bank group to another, and the originating branch has closed. To keep consistency with the sample definition, only moves that occur within the same province are considered. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix Table A3. DiD estimates for $\beta$, within the same bank

| | Credit indicator | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Inflow × Post | 0.021*** | 0.022*** | 0.021*** | 0.023*** |
| | (0.006) | (0.005) | (0.006) | (0.005) |
| $R^2$ | 0.802 | 0.806 | 0.803 | 0.807 |
| Observations | 23,988,940 | 23,988,940 | 23,988,940 | 23,988,940 |
| Branch-Firm fixed effects | ✓ | ✓ | ✓ | ✓ |
| Branch-Time fixed effects | | ✓ | | ✓ |
| Firm-Time fixed effects | | | ✓ | ✓ |

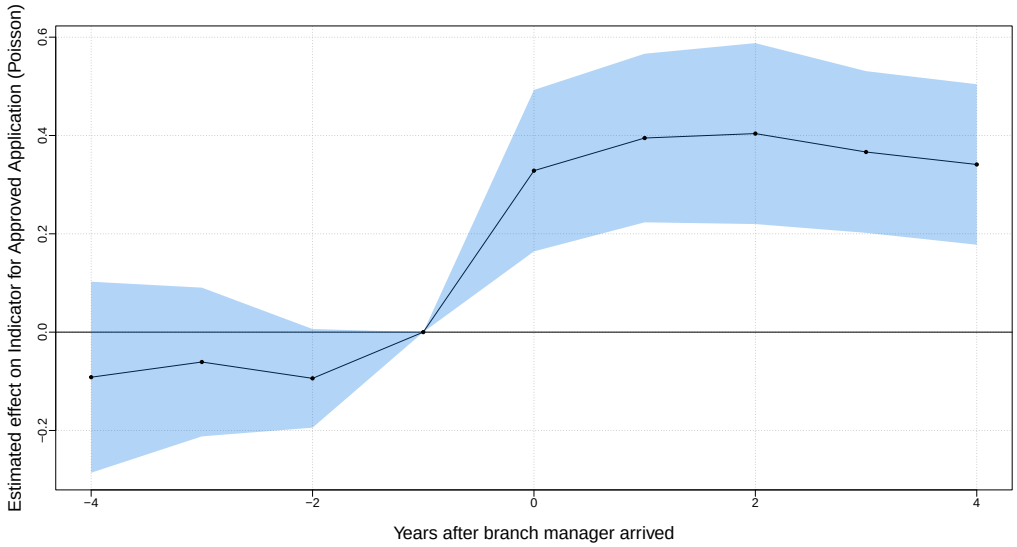
*Notes:* Estimates for the probability of a firm receiving credit from the branch where its previous loan officer has moved, within the same bank group. Column (1) includes only branch fixed effects, column (2) includes branch and time fixed effects, column (3) includes branch and firm fixed effects, and column (4) includes all fixed effects. Standard errors are clustered at the bank group - firm level. The sample includes all potential firm - branch pairs within a province, for the period 2009-2018. Only firms and branches that are present in the data in all years are included. Moves are defined as relocations of managers within the same bank group. To keep consistency with the sample definition, only moves that occur within the same province are considered. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix Figure A3. Increase in probability of applying for credit to the new bank: Poisson regression



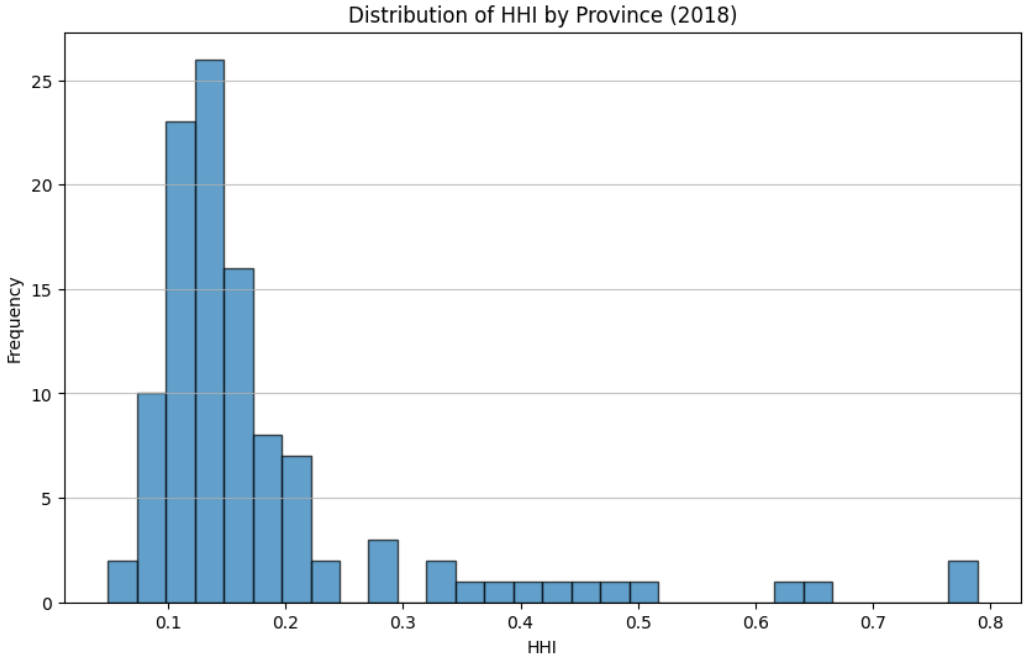

*Notes:* Event study estimates for the probability of a firm applying for credit in the bank where its bank manager has moved. Plot shows Poisson regression estimates and 95% confidence intervals for the coefficients $\beta_\tau$ in equation (3). Bank group - firm, bank group - time, and firm - time fixed effects are included. Standard errors are clustered at the bank group - firm level. Sample includes all potential firm - bank pairs, for the period 2009-2018. A pair is considered potential if in the province where the firm was present, there was at least one branch of the bank. Only firms and branches that are present in the data in all years are included. Moves are defined as relocations of managers from one bank group to another. To keep consistency with the sample definition, only moves that occur within the same province are considered. Estimates are obtained using the Sun and Abraham (2021) estimator.

Appendix Figure A4. Increase in probability of being approved for credit in the new bank: Poisson regression




*Notes:* Event study estimates for the probability of a firm being approved for credit in the bank where its bank manager has moved. Plot shows Poisson regression estimates and 95% confidence intervals for the coefficients $\beta_\tau$ in equation (4). Bank group - firm, bank group - time, and firm - time fixed effects are included. Standard errors are clustered at the bank group - firm level. Sample includes all potential firm - bank pairs, for the period 2009-2018. A pair is considered potential if in the province where the firm was present, there was at least one branch of the bank. Only firms and branches that are present in the data in all years are included. Moves are defined as relocations of managers from one bank group to another. To keep consistency with the sample definition, only moves that occur within the same province are considered. Estimates are obtained using the Sun and Abraham (2021) estimator.

Appendix Figure A5. Distribution of HHI by province in 2018




*Notes:* The figure plots the distribution of the Herfindahl-Hirschman Index (HHI) by province in 2018. The HHI is computed at the province level, using as weights the total amount of loans that each bank group has in that province. Provinces where the HHI is above 0.25 are considered to be highly concentrated markets, while provinces where the HHI is below 0.1 are considered to be too fragmented.

# C Interest rates breakdown

Appendix Table A4. Interest rates, within switchers comparison

| | Avg rate (2y) | Avg self-liquidating rate (2y) | Avg credit line rate (2y) | Avg rate, term loans (2y) |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Manager inflow | -0.505* | 0.069 | -0.641 | 0.111 |
| | (0.264) | (0.582) | (1.87) | (0.188) |
| Log. avg credit (2y) | -0.555*** | | | |
| | (0.019) | | | |
| Log. avg self-liquidating credit (2y) | | -0.302*** | | |
| | | (0.008) | | |
| Log. avg credit line credit (2y) | | | -1.00*** | |
| | | | (0.036) | |
| Log. avg credit, term loans (2y) | | | | -0.117*** |
| | | | | (0.006) |
| $R^2$ | 0.198 | 0.249 | 0.161 | 0.313 |
| Observations | 6,643 | 4,562 | 4,559 | 4,420 |
| Dependent variable mean | 2.55 | 4.93 | 11.5 | 3.07 |

*Notes:* Estimates for the effect on interest rates of switching with the loan officer. The dependent variable is the average interest rate on loans over a two-year period. The key independent variable is an indicator equal to one if the firm switched with the loan officer. All regressions include as controls the log of average credit granted over a two-year period, firm characteristics (age, size, riskiness), bank group, year and banker fixed effects. Standard errors are clustered at the firm level. The treated group is defined by the loans that switchers receive having followed their loan officer to a new bank. The control group is made of all other loans of switchers (before and after the switch). Column (1) estimates the effect of switching on the average interest rate. Columns (2) to (4) break down the effect by loan types, respectively self-liquidating loans (2), credit lines (3) or term loans (4). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix Table A5. Interest rates, switchers vs. portfolio firms (new rel.)

| | Avg rate (2y) | Avg self-liquidating rate (2y) | Avg credit line rate (2y) | Avg rate, term loans (2y) |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Manager inflow | -0.657 | 0.956 | -6.34*** | 0.529*** |
| | (0.699) | (0.733) | (2.30) | (0.151) |
| Log. avg credit (2y) | -0.669*** | | | |
| | (0.004) | | | |
| Log. avg self-liquidating credit (2y) | | -0.366*** | | |
| | | (0.002) | | |
| Log. avg credit line credit (2y) | | | -0.896*** | |
| | | | (0.014) | |
| Log. avg credit, term loans (2y) | | | | -0.130*** |
| | | | | (0.002) |
| $R^2$ | 0.197 | 0.221 | 0.090 | 0.334 |
| Observations | 23,609 | 15,355 | 14,654 | 15,955 |
| Dependent variable mean | 2.53 | 4.88 | 11.6 | 2.72 |

*Notes:* Estimates for the effect on interest rates of switching with the loan officer. The dependent variable is the average interest rate on loans over a two-year period. The key independent variable is an indicator equal to one if the firm switched with the loan officer. All regressions include as controls the log of average credit granted over a two-year period, firm characteristics (age, size, riskiness), bank group, year and banker fixed effects. Standard errors are clustered at the firm level. The treated group is defined by the loans that switchers receive having followed their loan officer to a new bank. What varies across columns is the definition of the control group. The control group is made of all the new relationships of portfolio firms, that did not switch with the banker. Column (1) estimates the effect of switching on the average interest rate. Columns (2) to (4) break down the effect by loan types, respectively self-liquidating loans (2), credit lines (3) or term loans (4). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix Table A6. Interest rates, switchers vs. portfolio firms (old rel.)

| | Avg rate (2y) | Avg self-liquidating rate (2y) | Avg credit line rate (2y) | Avg rate, term loans (2y) |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Switcher | -0.289* | -0.063 | -0.145 | 0.088 |
| | (0.172) | (0.320) | (0.788) | (0.143) |
| Log. avg credit (2y) | -0.716*** | | | |
| | (0.003) | | | |
| Log. avg self-liquidating credit (2y) | | -0.348*** | | |
| | | (0.002) | | |
| Log. avg credit line credit (2y) | | | -0.940*** | |
| | | | (0.005) | |
| Log. avg credit, term loans (2y) | | | | -0.096*** |
| | | | | (0.001) |
| $R^2$ | 0.188 | 0.222 | 0.112 | 0.187 |
| Observations | 35,585 | 24,317 | 26,483 | 22,209 |
| Dependent variable mean | 2.92 | 5.21 | 11.6 | 3.43 |

*Notes:* Estimates for the effect on interest rates of switching with the loan officer. The dependent variable is the average interest rate on loans over a two-year period. The key independent variable is an indicator equal to one if the firm switched with the loan officer. All regressions include as controls the log of average credit granted over a two-year period, firm characteristics (age, size, riskiness), bank group, year and banker fixed effects. Standard errors are clustered at the firm level. The treated group is defined by the loans that switchers receive having followed their loan officer to a new bank. The control group is made of all the old relationships of portfolio firms, that did not switch with the banker (i.e., relationships that existed before the banker moved to a new bank). Column (1) estimates the effect of switching on the average interest rate. Columns (2) to (4) break down the effect by loan types, respectively self-liquidating loans (2), credit lines (3) or term loans (4). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix Table A7. Interest rates, switchers vs. new branch

| | Avg rate (2y) | Avg self-liquidating rate (2y) | Avg credit line rate (2y) | Avg rate, term loans (2y) |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Manager inflow | 0.050 | -0.148 | -0.171 | 0.065 |
| | (0.144) | (0.269) | (0.594) | (0.125) |
| Log. avg credit (2y) | -0.268*** | | | |
| | (0.004) | | | |
| Log. avg self-liquidating credit (2y) | | -0.037*** | | |
| | | (0.0004) | | |
| Log. avg credit line credit (2y) | | | -0.152*** | |
| | | | (0.002) | |
| Log. avg credit, term loans (2y) | | | | -0.065*** |
| | | | | (0.0003) |
| $R^2$ | 0.123 | 0.166 | 0.042 | 0.271 |
| Observations | 68,555 | 43,288 | 41,535 | 49,168 |
| Dependent variable mean | 2.60 | 5.34 | 12.2 | 3.15 |

*Notes:* Estimates for the effect on interest rates of switching with the loan officer. The dependent variable is the average interest rate on loans over a two-year period. The key independent variable is an indicator equal to one if the firm switched with the loan officer. All regressions include as controls the log of average credit granted over a two-year period, firm characteristics (age, size, riskiness), bank group, year and banker fixed effects. Standard errors are clustered at the firm level. The treated group is defined by the loans that switchers receive having followed their loan officer to a new bank. The control group is made of all the new relationships of the new branch of the loan officer, that were not switchers. Column (1) estimates the effect of switching on the average interest rate. Columns (2) to (4) break down the effect by loan types, respectively self-liquidating loans (2), credit lines (3) or term loans (4). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix Table A8. Interest rates, switchers v. old relationships

| | Avg rate (2y) | Avg self-liquidating rate (2y) | Avg credit line rate (2y) | Avg rate, term loans (2y) |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Manager inflow | -0.926*** | -0.780** | -3.85* | -0.024 |
| | (0.195) | (0.339) | (2.28) | (0.289) |
| Log. avg credit (2y) | -0.773*** | | | |
| | (0.097) | | | |
| Log. avg self-liquidating credit (2y) | | -0.213*** | | |
| | | (0.025) | | |
| Log. avg credit line credit (2y) | | | -1.13*** | |
| | | | (0.224) | |
| Log. avg credit, term loans (2y) | | | | -0.053 |
| | | | | (0.043) |
| $R^2$ | 0.311 | 0.408 | 0.281 | 0.386 |
| Observations | 1,387 | 930 | 1,008 | 906 |
| Dependent variable mean | 2.93 | 5.44 | 11.9 | 3.34 |

*Notes:* Estimates for the effect on interest rates of switching with the loan officer. The dependent variable is the average interest rate on loans over a two-year period. The key independent variable is an indicator equal to one if the firm switched with the loan officer. All regressions include as controls the log of average credit granted over a two-year period, firm characteristics (age, size, riskiness), bank group, year and banker fixed effects. Standard errors are clustered at the firm level. The treated group is defined by the loans that switchers receive having followed their loan officer to a new bank. The control group is made of the loans the switchers had with the previous bank of the manager. Column (1) estimates the effect of switching on the average interest rate. Columns (2) to (4) break down the effect by loan types, respectively self-liquidating loans (2), credit lines (3) or term loans (4). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

# D Npl additional tables

Appendix Table A9. NPL probability, within switchers comparison

| | Npl probability | | |
|---|---|---|---|
| | (0 years) | (1 year) | (2 years) |
| | (1) | (2) | (3) |
| Switcher | -0.014** | -0.007 | 0.016 |
| | (0.006) | (0.011) | (0.020) |
| Log. avg | | | |
| self-liquidating credit (2y) | -0.0009*** | -0.0006** | -0.0006 |
| | (0.0003) | (0.0003) | (0.0004) |
| Log. avg | | | |
| credit line credit (2y) | $8.82 \times 10^{-5}$ | 0.0002 | 0.0005 |
| | (0.0004) | (0.0005) | (0.0005) |
| Log. avg | | | |
| credit, term loans (2y) | 0.0007*** | 0.001*** | 0.002*** |
| | (0.0002) | (0.0003) | (0.0004) |
| $R^2$ | 0.115 | 0.125 | 0.128 |
| Observations | 13,320 | 12,604 | 11,413 |
| Dependent variable mean | 0.017 | 0.023 | 0.030 |

*Notes:* Estimates for the effect on NPL probability of switching with the loan officer. The dependent variable is the probability of non-performing loans (NPL) over a two-year period. The key independent variable is an indicator equal to one if the firm switched with the loan officer. All regressions include as controls the log of credit granted (broken down by type of loan), firm characteristics (age, size, riskiness), municipality, bank group-year and banker fixed effects. Standard errors are clustered at the firm level. The treated group is defined by the loans that switchers receive having followed their loan officer to a new bank. The control group is made of all other loans of switchers (before and after the switch). Columns (1) to (3) show NPL probabilities in different time horizons. Respectively, the same year the loan has been given, one year after and two years after. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix Table A10. NPL probability, switchers vs. portfolio firms (new rel.)

| | Npl probability | | |
|---|---|---|---|
| | (0 years) | (1 year) | (2 years) |
| | (1) | (2) | (3) |
| Switcher | -0.011 | -0.003 | 0.012 |
| | (0.007) | (0.008) | (0.014) |
| Log. avg | | | |
| self-liquidating credit (2y) | -0.001*** | -0.001*** | -0.0009* |
| | (0.0003) | (0.0003) | (0.0005) |
| Log. avg | | | |
| credit line credit (2y) | -0.0002 | $3.98 \times 10^{-5}$ | 0.0006 |
| | (0.0005) | (0.0005) | (0.0006) |
| Log. avg | | | |
| credit, term loans (2y) | 0.0009*** | 0.001*** | 0.002*** |
| | (0.0001) | (0.0002) | (0.0003) |
| $R^2$ | 0.101 | 0.109 | 0.117 |
| Observations | 45,700 | 40,370 | 30,536 |
| Dependent variable mean | 0.027 | 0.035 | 0.043 |

*Notes:* Estimates for the effect on NPL probability of switching with the loan officer. The dependent variable is the probability of non-performing loans (NPL) over a two-year period. The key independent variable is an indicator equal to one if the firm switched with the loan officer. All regressions include as controls the log of credit granted (broken down by type of loan), firm characteristics (age, size, riskiness), municipality, bank group-year and banker fixed effects. Standard errors are clustered at the firm level. The treated group is defined by the loans that switchers receive having followed their loan officer to a new bank. The control group is made of all the new relationships of portfolio firms, that did not switch with the banker. Columns (1) to (3) show NPL probabilities in different time horizons. Respectively, the same year the loan has been given, one year after and two years after. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix Table A11. NPL probability, switchers vs. portfolio firms (old rel.)

| | Npl probability | | |
|---|---|---|---|
| | (0 years) | (1 year) | (2 years) |
| | (1) | (2) | (3) |
| Switcher | -0.020* | -0.017 | -0.015 |
| | (0.011) | (0.011) | (0.015) |
| Log. avg | | | |
| self-liquidating credit (2y) | -0.0003** | -0.0003 | -0.0002 |
| | (0.0002) | (0.0002) | (0.0003) |
| Log. avg | | | |
| credit line credit (2y) | -0.0002 | $6.14 \times 10^{-5}$ | 0.0003 |
| | (0.0003) | (0.0003) | (0.0004) |
| Log. avg | | | |
| credit, term loans (2y) | 0.0005*** | 0.0008*** | 0.001*** |
| | (0.0001) | (0.0002) | (0.0002) |
| $R^2$ | 0.055 | 0.061 | 0.066 |
| Observations | 65,195 | 65,140 | 64,991 |
| Dependent variable mean | 0.015 | 0.023 | 0.032 |

*Notes:* Estimates for the effect on NPL probability of switching with the loan officer. The dependent variable is the probability of non-performing loans (NPL) over a two-year period. The key independent variable is an indicator equal to one if the firm switched with the loan officer. All regressions include as controls the log of credit granted (broken down by type of loan), firm characteristics (age, size, riskiness), municipality, bank group-year and banker fixed effects. Standard errors are clustered at the firm level. The treated group is defined by the loans that switchers receive having followed their loan officer to a new bank. The control group is made of all the old relationships of portfolio firms, that did not switch with the banker (i.e., relationships that existed before the banker moved to a new bank). Columns (1) to (3) show NPL probabilities in different time horizons. Respectively, the same year the loan has been given, one year after and two years after. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix Table A12. NPL probability, switchers vs. new branch

| | Npl probability | | |
|---|---|---|---|
| | (0 years) | (1 year) | (2 years) |
| | (1) | (2) | (3) |
| Switcher | -0.043** | -0.043*** | -0.030*** |
| | (0.019) | (0.008) | (0.008) |
| Log. avg | | | |
| self-liquidating credit (2y) | -0.001*** | -0.0005*** | 0.0001 |
| | (0.0001) | $(4.88 \times 10^{-5})$ | (0.0001) |
| Log. avg | | | |
| credit line credit (2y) | 0.0005** | 0.0006** | 0.001*** |
| | (0.0002) | (0.0002) | (0.0002) |
| Log. avg | | | |
| credit, term loans (2y) | 0.0003** | 0.0008*** | 0.001*** |
| | (0.0001) | $(5.37 \times 10^{-5})$ | $(4.37 \times 10^{-5})$ |
| $R^2$ | 0.048 | 0.042 | 0.041 |
| Observations | 187,389 | 167,876 | 142,976 |
| Dependent variable mean | 0.016 | 0.025 | 0.032 |

*Notes:* Estimates for the effect on NPL probability of switching with the loan officer. The dependent variable is the probability of non-performing loans (NPL) over a two-year period. The key independent variable is an indicator equal to one if the firm switched with the loan officer. All regressions include as controls the log of credit granted (broken down by type of loan), firm characteristics (age, size, riskiness), municipality, bank group-year and banker fixed effects. Standard errors are clustered at the firm level. The treated group is defined by the loans that switchers receive having followed their loan officer to a new bank. The control group is made of all the new relationships of the new branch of the loan officer, that were not switchers. Columns (1) to (3) show NPL probabilities in different time horizons. Respectively, the same year the loan has been given, one year after and two years after. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix Table A13. NPL probability performance for risky firms

| | NPL probability | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Switcher | -0.015** | -0.008 | -0.021** | -0.043** |
| | (0.006) | (0.007) | (0.011) | (0.019) |
| Switcher × Risky | 0.011 | -0.037** | 0.012 | -0.0007 |
| | (0.015) | (0.017) | (0.014) | (0.011) |
| $R^2$ | 0.115 | 0.101 | 0.055 | 0.048 |
| Observations | 13,320 | 45,700 | 65,195 | 187,389 |
| Dependent variable mean | 0.017 | 0.027 | 0.015 | 0.016 |

*Notes:* Estimates for the effect on NPL probability of switching with the loan officer. The dependent variable is the probability of non-performing loans (NPL) over a two-year period. The key independent variable is the interaction between the indicator equal to one if the firm switched with the loan officer and a dummy equal to one if the firm is categorized as risky by Cebi-Cerved (i.e., it has a risk score of 7 or higher on a scale from 1 to 10). All regressions include as controls the log of average credit granted over a two-year period, firm characteristics (age, size, riskiness), bank, year and banker fixed effects. Standard errors are clustered at the firm level. In all columns, the treated group is defined by the loans that switchers receive having followed their loan officer to a new bank. What varies across columns is the definition of the control group. In column (1) the control group is made of all other loans of switchers (before and after the switch). In column (2) the control group is made of all the new relationships of portfolio firms, that did not switch with the banker. In column (3) the control group is made of all the old relationships of portfolio firms, that did not switch with the banker (i.e., relationships that existed before the banker moved to a new bank). In column (4) the control group is made of all the new relationships of the new branch of the loan officer, that were not switchers. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

# E Credit amounts: loan-type breakdowns

This appendix reports, for the saturated specification, the by-loan-type breakdowns behind the main-text Tables 13 and 14. Each table splits the total-credit effect into self-liquidating credit, credit lines and term loans; the breakdown columns are restricted to observations with positive credit of that type.

## E.1 Movers, before vs after (Table 13)

Appendix Table A14. New vs old branch relationship, by loan type

| | Log credit granted | | | |
|---|---|---|---|---|
| | Total | Self-liquidating | Credit lines | Term |
| | (1) | (2) | (3) | (4) |
| After move | -0.070* | -0.092** | -0.147*** | -0.113 |
| | (0.037) | (0.038) | (0.040) | (0.084) |
| $R^2$ | 0.926 | 0.910 | 0.882 | 0.922 |
| Observations | 1,486 | 932 | 1,124 | 608 |
| Dependent variable mean | 12.8 | 11.9 | 10.5 | 12.6 |

*Notes:* Saturated specification (pair (firm × transition) fixed effects and firm characteristics), by loan type. Dependent variable: log of credit granted at the old branch (before the move) vs at the new branch the manager opens (after). Breakdown columns are restricted to observations with positive credit of that type. Standard errors clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix Table A15. Firm total credit around the move, by loan type

| | Log credit granted | | | |
|---|---|---|---|---|
| | Total | Self-liquidating | Credit lines | Term |
| | (1) | (2) | (3) | (4) |
| After move | 0.626*** | 0.169*** | 0.184*** | 0.155*** |
| | (0.087) | (0.021) | (0.025) | (0.047) |
| $R^2$ | 0.853 | 0.987 | 0.982 | 0.962 |
| Observations | 1,482 | 1,108 | 1,308 | 1,124 |
| Dependent variable mean | 13.8 | 13.4 | 12.0 | 13.6 |

*Notes:* Saturated specification (pair (firm × transition) fixed effects and firm characteristics), by loan type. Dependent variable: log of the firm's total credit across all banks and branches, at the year before vs the year of the move. Breakdown columns are restricted to observations with positive credit of that type. Standard errors clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix Table A16. Old-branch retention around the move, by loan type

| | Pr(credit remains at old branch) | Self-liquidating | Credit lines | Term |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| After move | | -0.146*** | -0.111** | -0.184*** |
| | | (0.052) | (0.055) | (0.068) |
| $R^2$ | | 0.949 | 0.931 | 0.967 |
| Observations | | 378 | 454 | 394 |
| Dependent variable mean | | 12.7 | 11.0 | 13.1 |

*Notes:* Saturated specification (pair (firm × transition) fixed effects and firm characteristics), by loan type. In column (1) the dependent variable is an indicator that any credit remains at the old branch after the move (*coefficient pending re-estimation on the current data*); in columns (2)–(4) it is the log of credit of that type still held at the old branch, restricted to observations with positive credit. Standard errors clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

### E.2 Followed relationships, cross-section (Table 14)

Appendix Table A17. Followed relationship vs the follower firm's own book, by loan type

| | Log credit granted | | | |
|---|---|---|---|---|
| | Total | Self-liquidating | Credit lines | Term |
| | (1) | (2) | (3) | (4) |
| Manager inflow | -0.401* | -0.583 | -0.294 | -0.120 |
| | (0.230) | (0.393) | (0.384) | (0.350) |
| $R^2$ | 0.562 | 0.488 | 0.488 | 0.500 |
| Observations | 12,690 | 7,326 | 8,447 | 8,237 |
| Dependent variable mean | 13.5 | 12.8 | 11.4 | 13.1 |

*Notes:* Saturated specification (municipality, bank group-year and manager fixed effects, plus firm characteristics), by loan type. The control group is the follower firm's own other relationships (any year). Breakdown columns are restricted to observations with positive credit of that type. Standard errors clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix Table A18. Followed relationship vs portfolio firms' other new relationships, by loan type

| | Log credit granted | | | |
|---|---|---|---|---|
| | Total | Self-liquidating | Credit lines | Term |
| | (1) | (2) | (3) | (4) |
| Manager inflow | 0.556* | -0.404 | 0.084 | 0.172 |
| | (0.315) | (0.699) | (0.528) | (0.552) |
| $R^2$ | 0.465 | 0.396 | 0.327 | 0.395 |
| Observations | 44,795 | 25,276 | 28,541 | 29,508 |
| Dependent variable mean | 13.1 | 12.7 | 11.0 | 12.7 |

*Notes:* Saturated specification (municipality, bank group-year and manager fixed effects, plus firm characteristics), by loan type. The control group is portfolio firms' other new relationships (formed at or after the move). Breakdown columns are restricted to observations with positive

credit of that type. Standard errors clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix Table A19. Followed relationship vs portfolio firms' old relationships, by loan type

| | Log credit granted | | | |
|---|---|---|---|---|
| | Total | Self-liquidating | Credit lines | Term |
| | (1) | (2) | (3) | (4) |
| Manager inflow | 0.002 | 0.011 | -0.136 | 0.011 |
| | (0.096) | (0.125) | (0.122) | (0.168) |
| $R^2$ | 0.395 | 0.406 | 0.303 | 0.300 |
| Observations | 64,275 | 38,166 | 46,837 | 39,319 |
| Dependent variable mean | 13.0 | 12.6 | 11.2 | 12.5 |

*Notes:* Saturated specification (municipality, bank group-year and manager fixed effects, plus firm characteristics), by loan type. The control group is portfolio firms' old, pre-move relationships. Breakdown columns are restricted to observations with positive credit of that type. Standard errors clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix Table A20. Followed relationship vs other new clients at the destination branch, by loan type

| | Log credit granted | | | |
|---|---|---|---|---|
| | Total | Self-liquidating | Credit lines | Term |
| | (1) | (2) | (3) | (4) |
| Manager inflow | -0.353* | -0.192 | 0.241 | -0.532** |
| | (0.196) | (0.429) | (0.658) | (0.265) |
| $R^2$ | 0.277 | 0.330 | 0.305 | 0.220 |
| Observations | 187,256 | 42,539 | 43,852 | 154,234 |
| Dependent variable mean | 12.3 | 12.6 | 10.8 | 12.1 |

*Notes:* Saturated specification (municipality, bank group-year and manager fixed effects, plus firm characteristics), by loan type. The control group is other new clients at the branches that

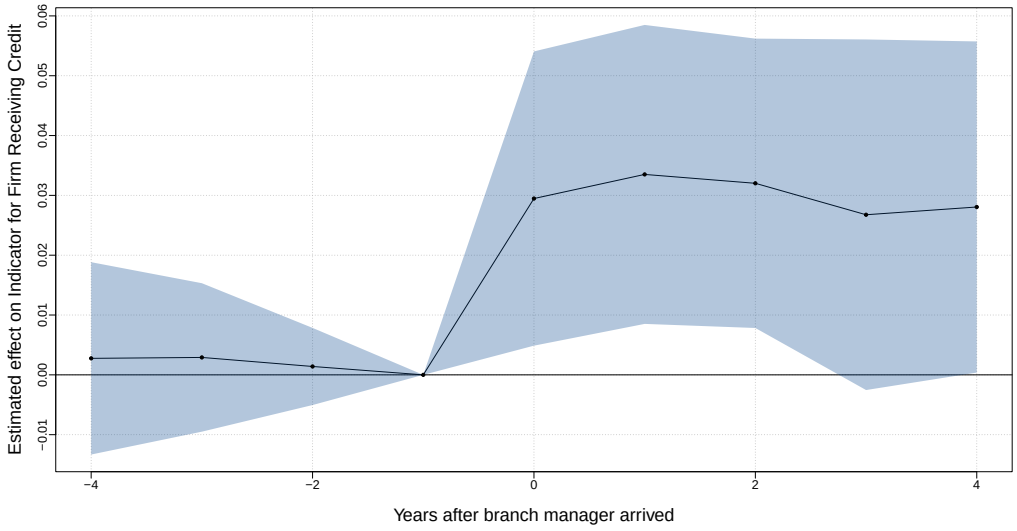
received a followed manager. Breakdown columns are restricted to observations with positive credit of that type. Standard errors clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

# F Robustness

## F.1 Robustness: branches with at most 100 firms

Figure A6 repeats the event study of Figure 2, restricting the sample of moves to managers leaving branches that had given credit to at most 100 unique firms per year in the two years before the move (instead of 150).

Appendix Figure A6. Increase in probability of giving credit to portfolio firms after the move: branches with at most 100 firms

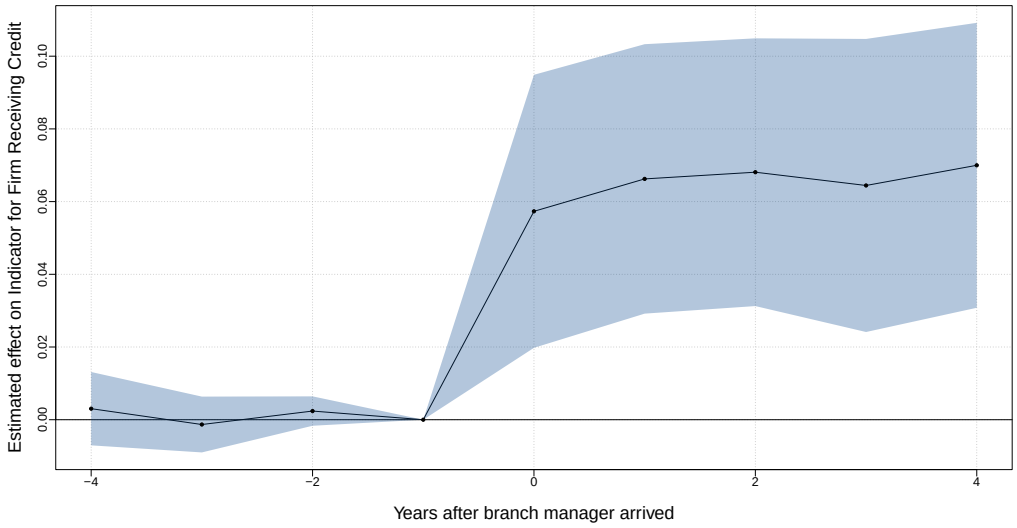



*Notes:* Event study estimates for the probability of a firm receiving credit from the branch where its previous loan officer has moved. Plot shows OLS estimates and 95% confidence intervals for the coefficients $\beta_\tau$ in equation (2). Only moves from branches that had given credit to at most 100 unique firms per year in the two years before the move are considered. Branch - firm, branch - time, and firm - time fixed effects are included. Standard errors are double-clustered at the bank group and firm level. Sample includes all potential firm - branch pairs within a province, for the period 2009-2018. Only firms and branches that are present in the data in all years are included. Moves are defined as relocations of managers from one bank group to another, within the same province. Estimates are obtained using the Sun and Abraham (2021) estimator.

### F.2 Robustness: branches with at most 50 firms

Figure A7 repeats the event study of Figure 2, restricting the sample of moves to managers leaving branches that had given credit to at most 50 unique firms per year in the two years before the move (instead of 150).

Appendix Figure A7. Increase in probability of giving credit to portfolio firms after the move: branches with at most 50 firms



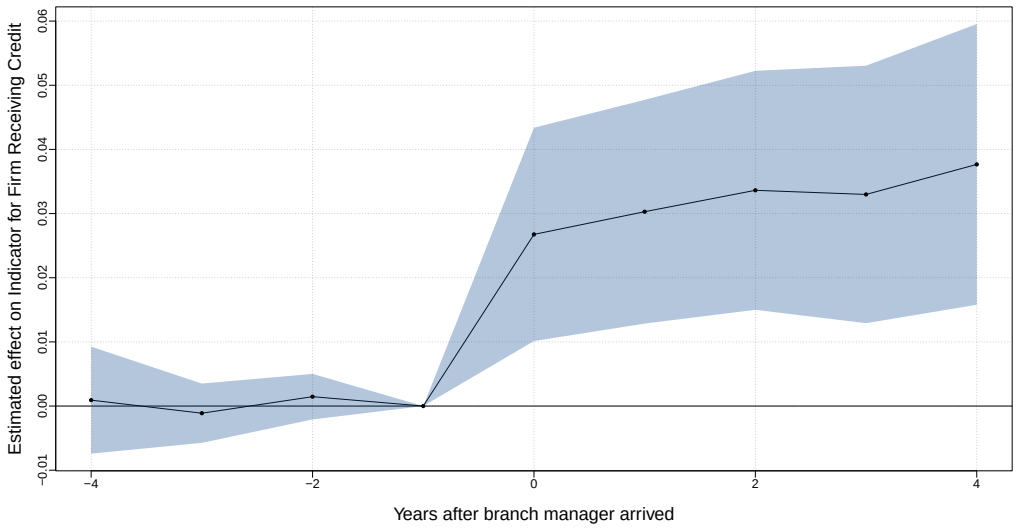

*Notes:* Event study estimates for the probability of a firm receiving credit from the branch where its previous loan officer has moved. Plot shows OLS estimates and 95% confidence intervals for the coefficients $\beta_\tau$ in equation (2). Only moves from branches that had given credit to at most 50 unique firms per year in the two years before the move are considered. Branch - firm, branch - time, and firm - time fixed effects are included. Standard errors are double-clustered at the bank group and firm level. Sample includes all potential firm - branch pairs within a province, for the period 2009-2018. Only firms and branches that are present in the data in all years are included. Moves are defined as relocations of managers from one bank group to another, within the same province. Estimates are obtained using the Sun and Abraham (2021) estimator.

### F.3 Robustness: aggregation at the bank group - province level

A firm may already have a credit relationship with the manager's new bank group through another branch in the same province. To address this concern, Figure A8 repeats the event study of Figure 2, aggregating observations at the firm - bank group - province level: the outcome is equal to one if the firm has credit with any branch of the bank group in the province, and the pair is treated from the year in which a manager who had the firm in her portfolio joins any branch of the bank group in the

province.

Appendix Figure A8. Increase in probability of giving credit to portfolio firms after the move: bank group - province level




*Notes:* Event study estimates for the probability of a firm receiving credit from the bank group, in the province, where its previous loan officer has moved. Plot shows OLS estimates and 95% confidence intervals for the coefficients $\beta_\tau$ in equation (2), estimated at the firm - bank group - province level. Firm - bank group - province, bank group - province - time, and firm - time fixed effects are included. Standard errors are double-clustered at the bank group and firm level. Moves are defined as relocations of managers from one bank group to another, within the same province, from branches that had given credit to at most 150 unique firms per year in the two years before the move. Estimates are obtained using the Sun and Abraham (2021) estimator.